# GWASH vs. LDSC Intercept Choice

GWASH, LDSC free intercept, and LDSC fixed intercept compared together, under real chr1/chr22 LD structure

## Executive Summary

This report compares GWASH's heritability estimator directly against the two standard LDSC intercept choices (free intercept and fixed intercept, both weighted - the default LDSC settings used as the main estimators throughout this project's companion reports). The free-intercept estimate uses LDSC's two-step procedure (its default when the intercept is free); the fixed-intercept estimate runs as a single joint regression pass, since two-step is only defined for a free intercept. The comparison spans every scale (n=500/1000/2000/5000), chromosome (chr1, chr22), and the project's four core scenarios (reference panel present or not, stratification vector f shared or drawn independently).

**Headline finding:** in the three well-specified scenarios (no reference panel with either f-sharing choice, and a reference panel with *shared* f), GWASH and both LDSC intercept choices track each other and the theory line closely across the $F_{ST}$ range shown here (0-0.1; per reviewer feedback, wider $F_{ST}$ values are no longer the focus). In the pathological scenario - a reference panel whose stratification vector f is mismatched with the main sample - all three estimators grow with sample size, but at very different rates. At $F_{ST}$=0.1, chr1, LDSC's free-intercept estimate rises from 0.28 (n=500) to 1.33 (n=5000) - a 4.7× increase - and the fixed-intercept estimate rises from 0.96 to 1.74 (1.8×). GWASH, over that same range of scales, rises only from 0.65 to 0.95 (1.5×) - growing, but far more slowly than LDSC's free-intercept estimate, and ending up below both LDSC variants at the largest scale despite starting above the free-intercept one at the smallest.

# Methods

See the companion reports (realistic_LD_sharedf_vs_indepf_summary.pdf and ldsc_estimator_variants_comparison.pdf) for full simulation methodology. The three estimators compared here, all computed from the same per-replicate data (same X, y, LD scores):

- **h2_gwash**: GWASH
- **h2_ldsc_reg**: LDSC, free intercept (weighted, two-step)
- **h2_ldsc_fixed**: LDSC, fixed intercept (weighted)

All panels use a linear y-axis. The grey dashed line is h2_pop=0.3, the true simulated genetic heritability. The black dotted line is the theoretical total h2 (h2_pop + $sigma_s^2$/C_f, C_f = mean($f^2$/(1+$f^2$))), using that scenario's own per-replicate f draws.

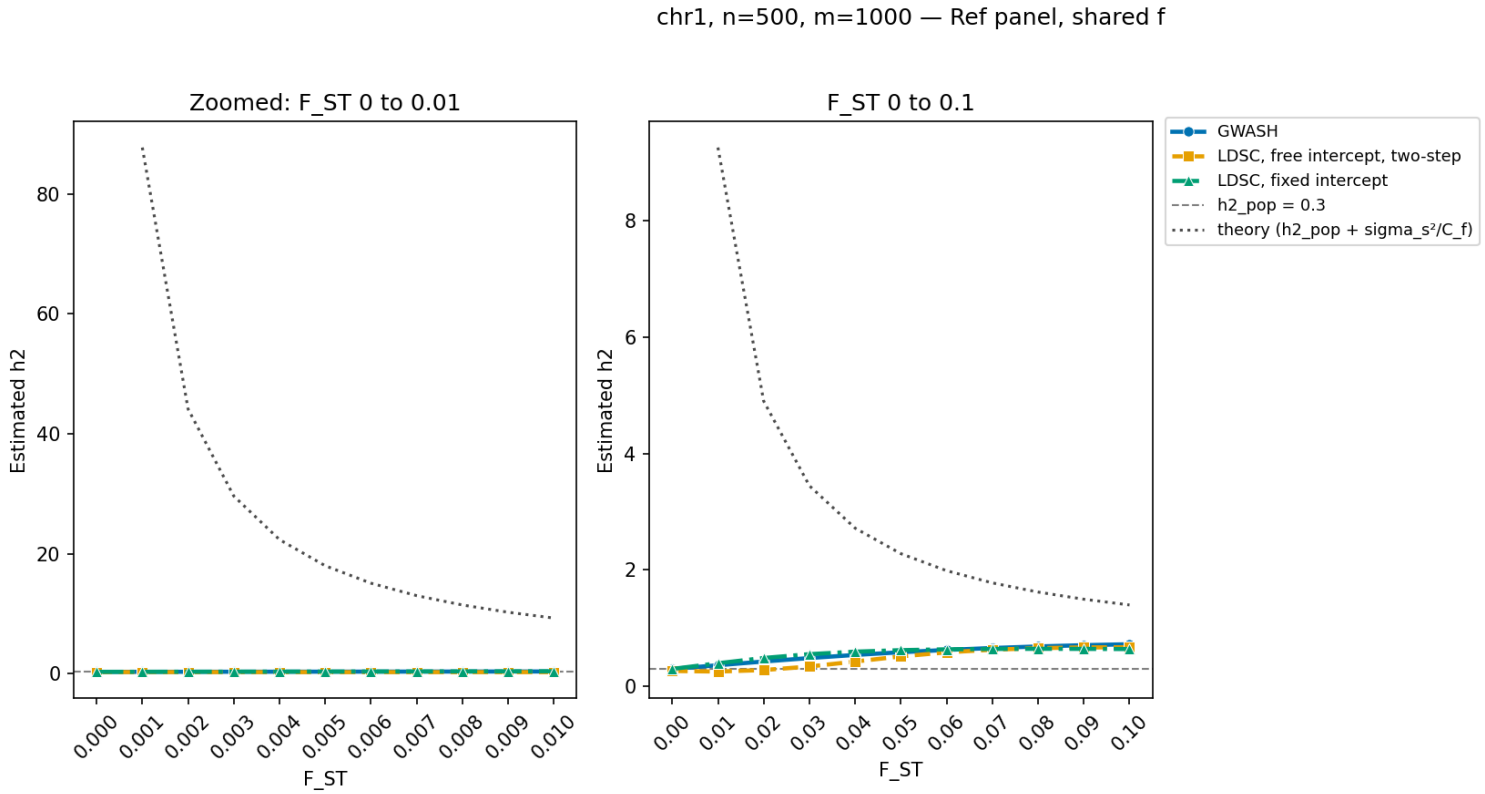
# Summary: GWASH vs. LDSC at F_ST=0.1, Reference Panel + Independent f

Mean h2 estimate (100 replicates) for each estimator at $F_{ST}$=0.1, the upper edge of the $F_{ST}$ range this report now focuses on (per reviewer feedback), in the scenario where the companion report found the largest divergence: a reference panel whose f does not match the main sample's.

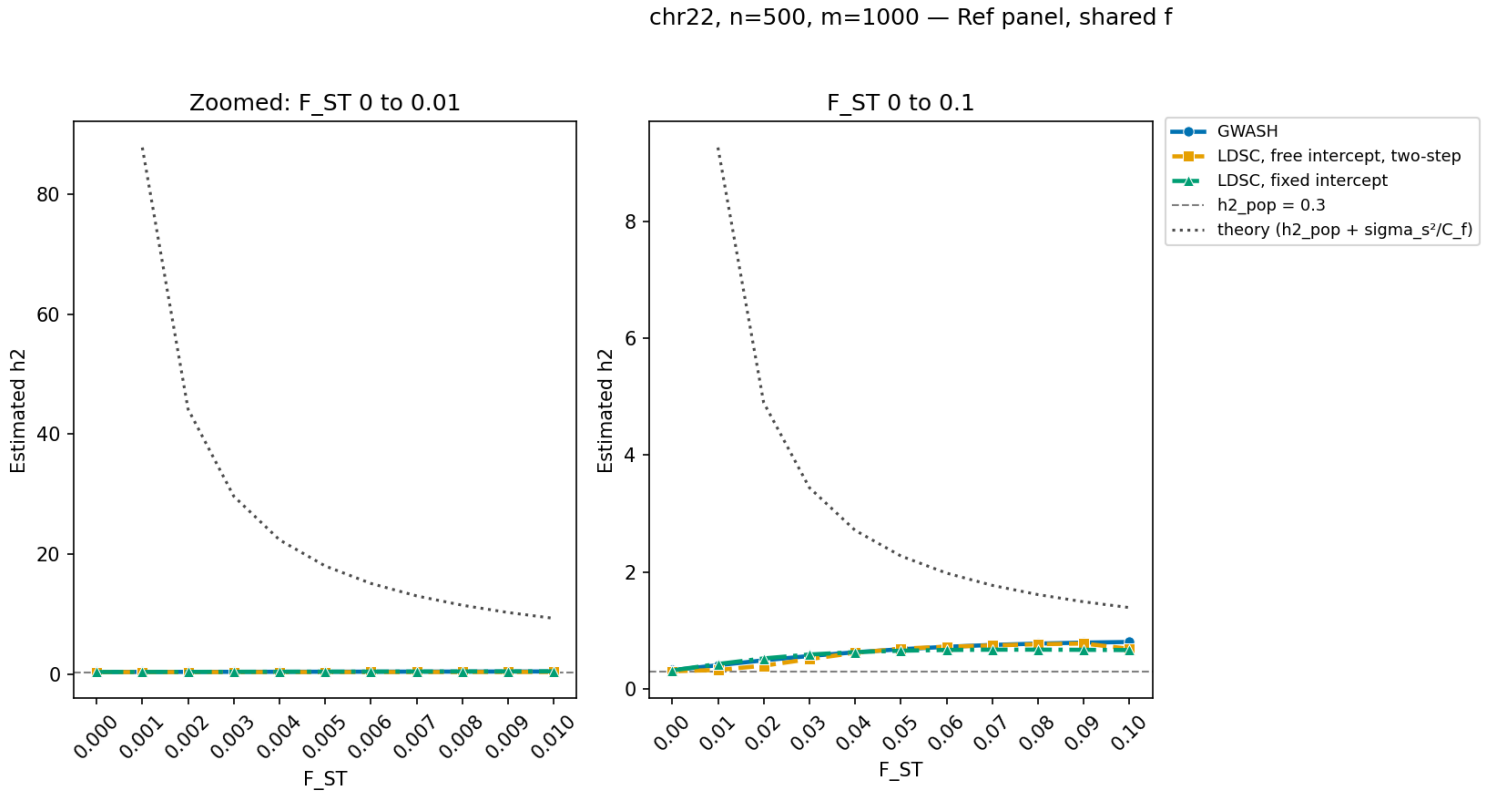
| Chrom | n | GWASH | LDSC free intercept | LDSC fixed intercept |
|---|---|---|---|---|
| chr1 | 500 | 0.647 | 0.284 | 0.960 |
| chr1 | 1000 | 0.734 | 0.438 | 1.123 |
| chr1 | 2000 | 0.791 | 0.735 | 1.340 |
| chr1 | 5000 | 0.949 | 1.325 | 1.743 |
| chr22 | 500 | 0.709 | 0.285 | 1.061 |
| chr22 | 1000 | 0.704 | 0.436 | 1.132 |
| chr22 | 2000 | 0.784 | 0.752 | 1.380 |
| chr22 | 5000 | 0.926 | 1.283 | 1.692 |

All three estimators grow with scale in this scenario, but at different rates: LDSC's free-intercept estimate grows fastest (4.5-4.7× from n=500 to n=5000), LDSC's fixed-intercept estimate grows more moderately (1.6-1.8×), and GWASH grows the least (1.3-1.5×), ending up the smallest of the three at n=5000 on both chromosomes despite starting above the free-intercept estimate at n=500. GWASH is still biased above h2_pop=0.3 throughout, but its sensitivity to the f-mismatch grows more slowly with sample size than either LDSC intercept choice.

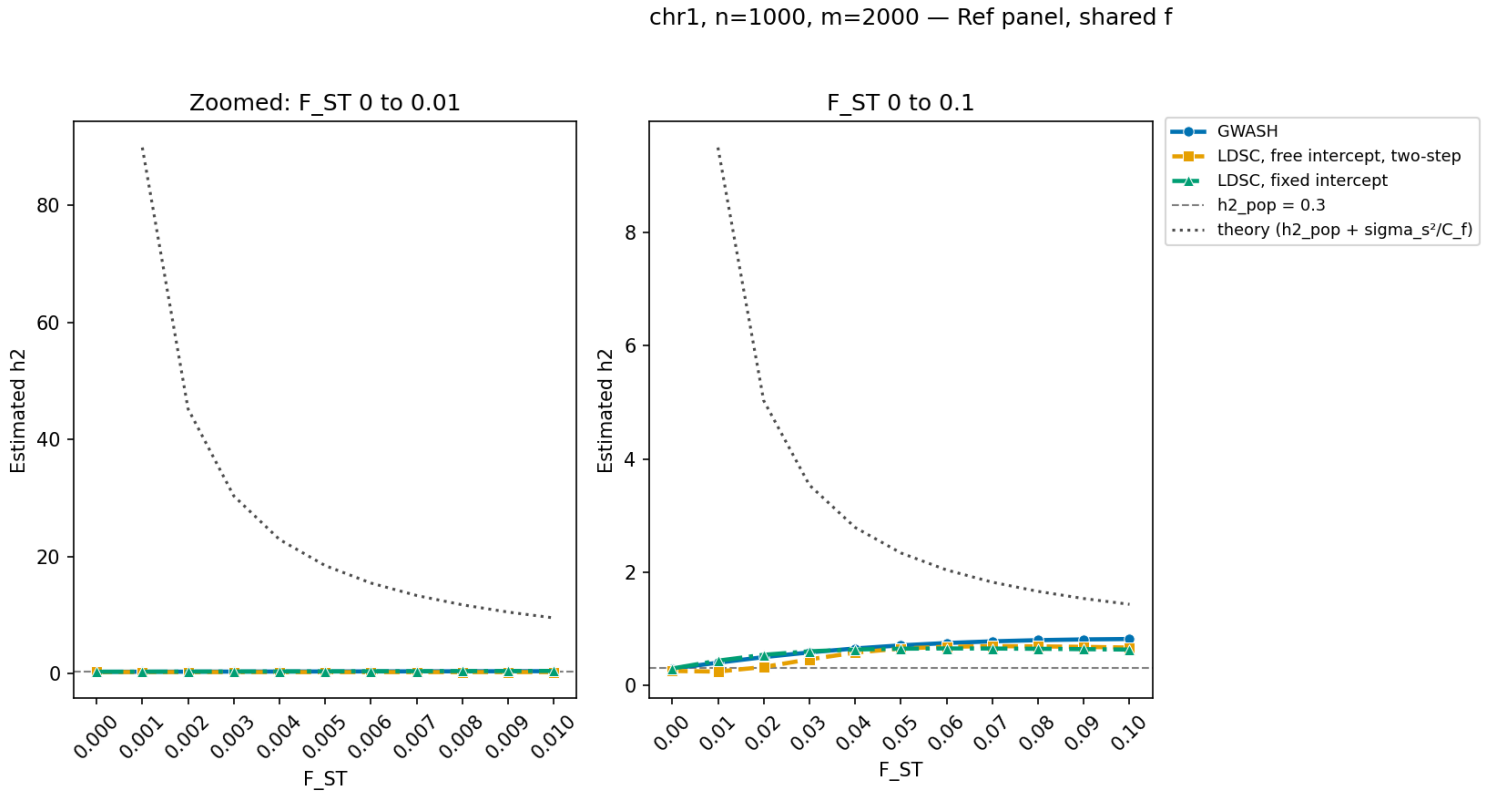
# Figures: Ref panel, shared f



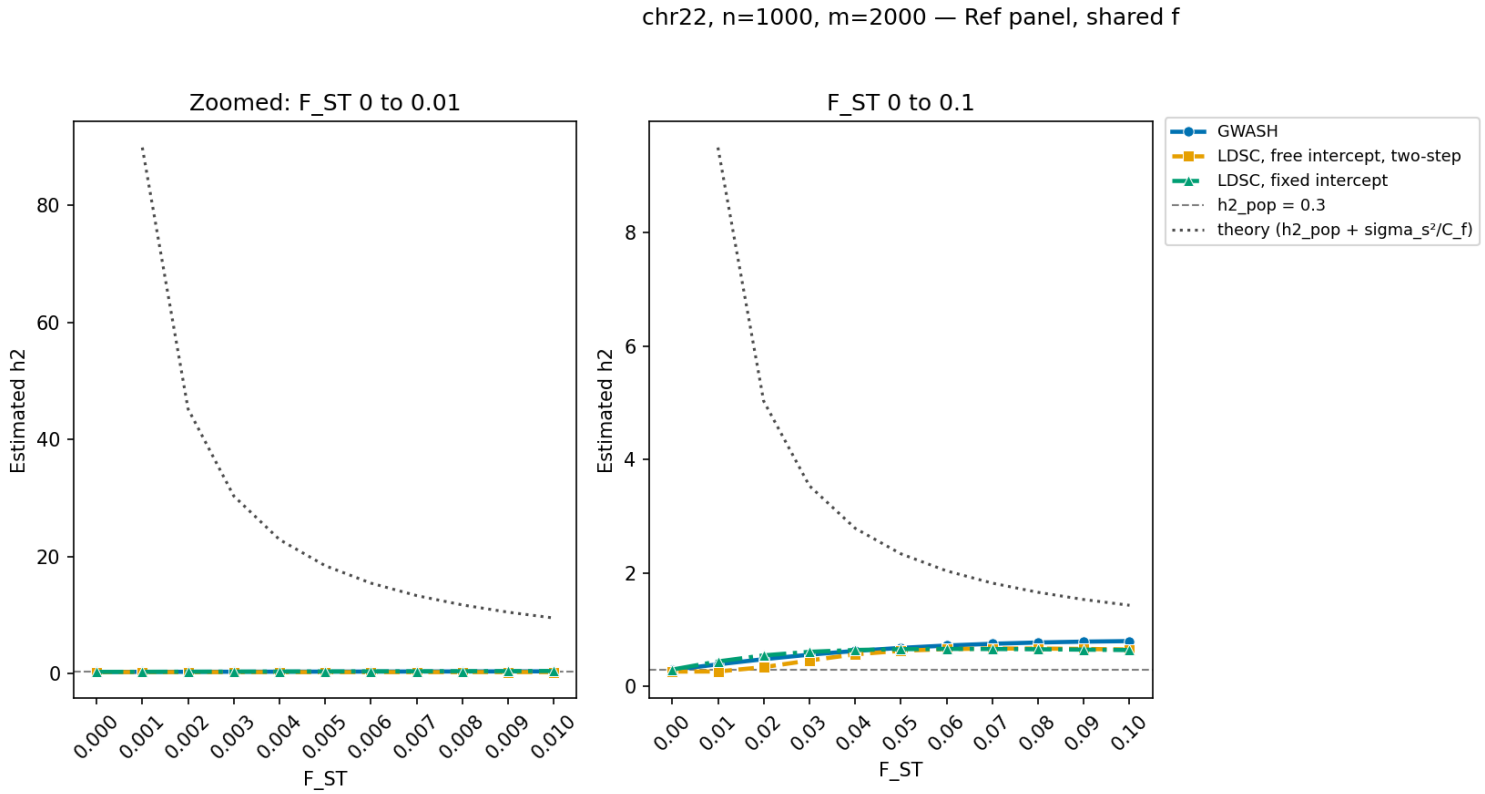
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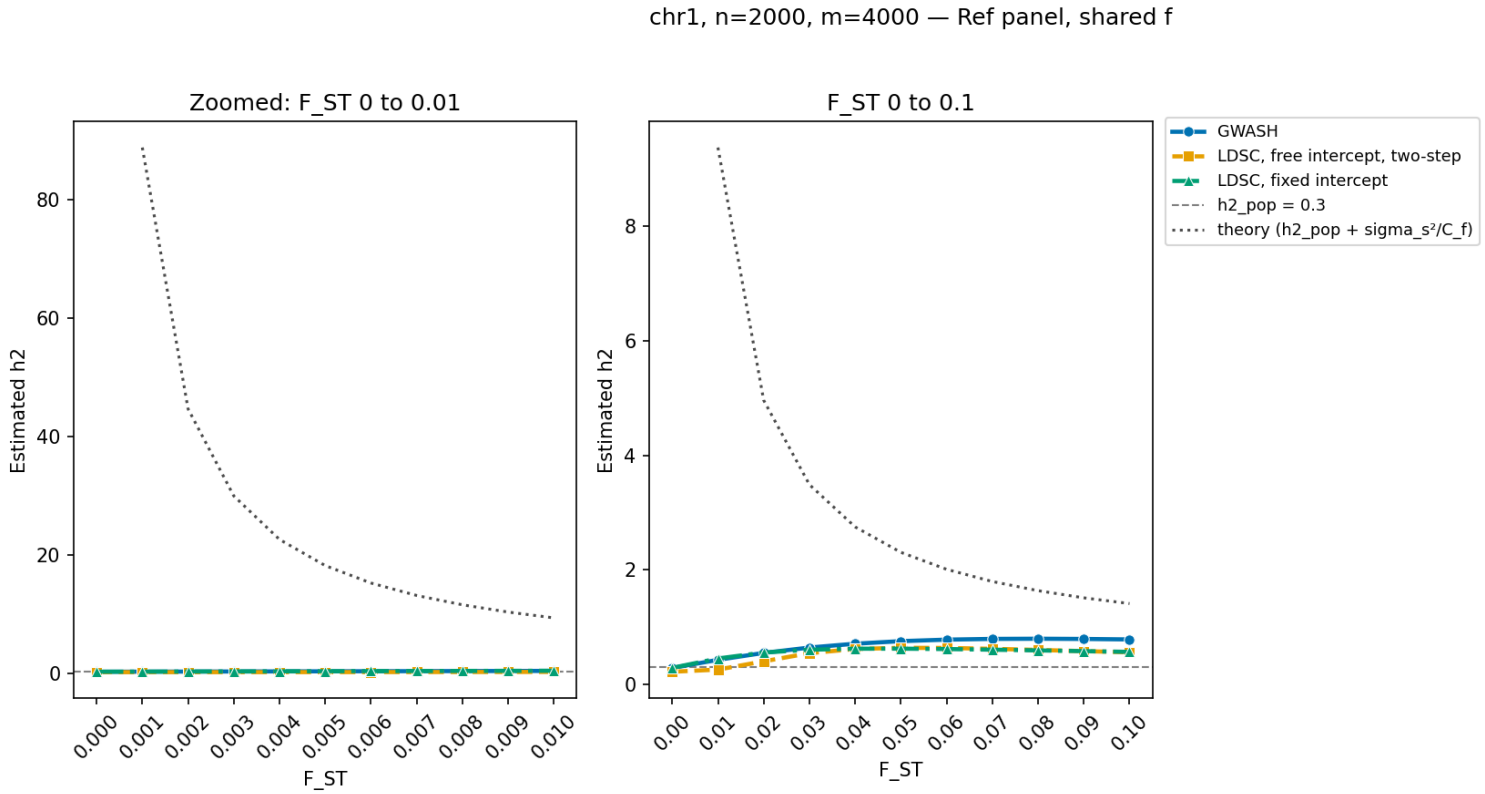
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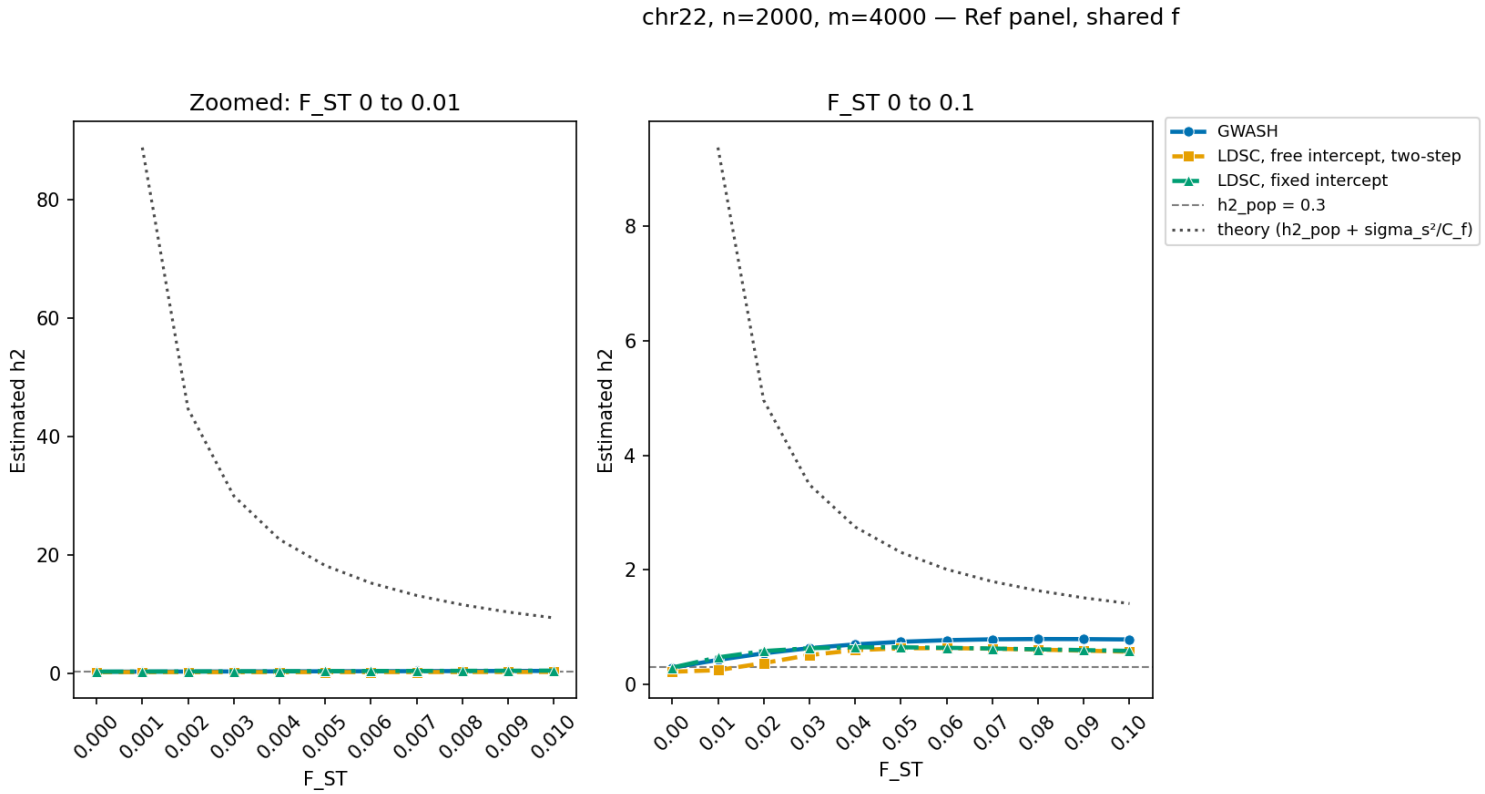
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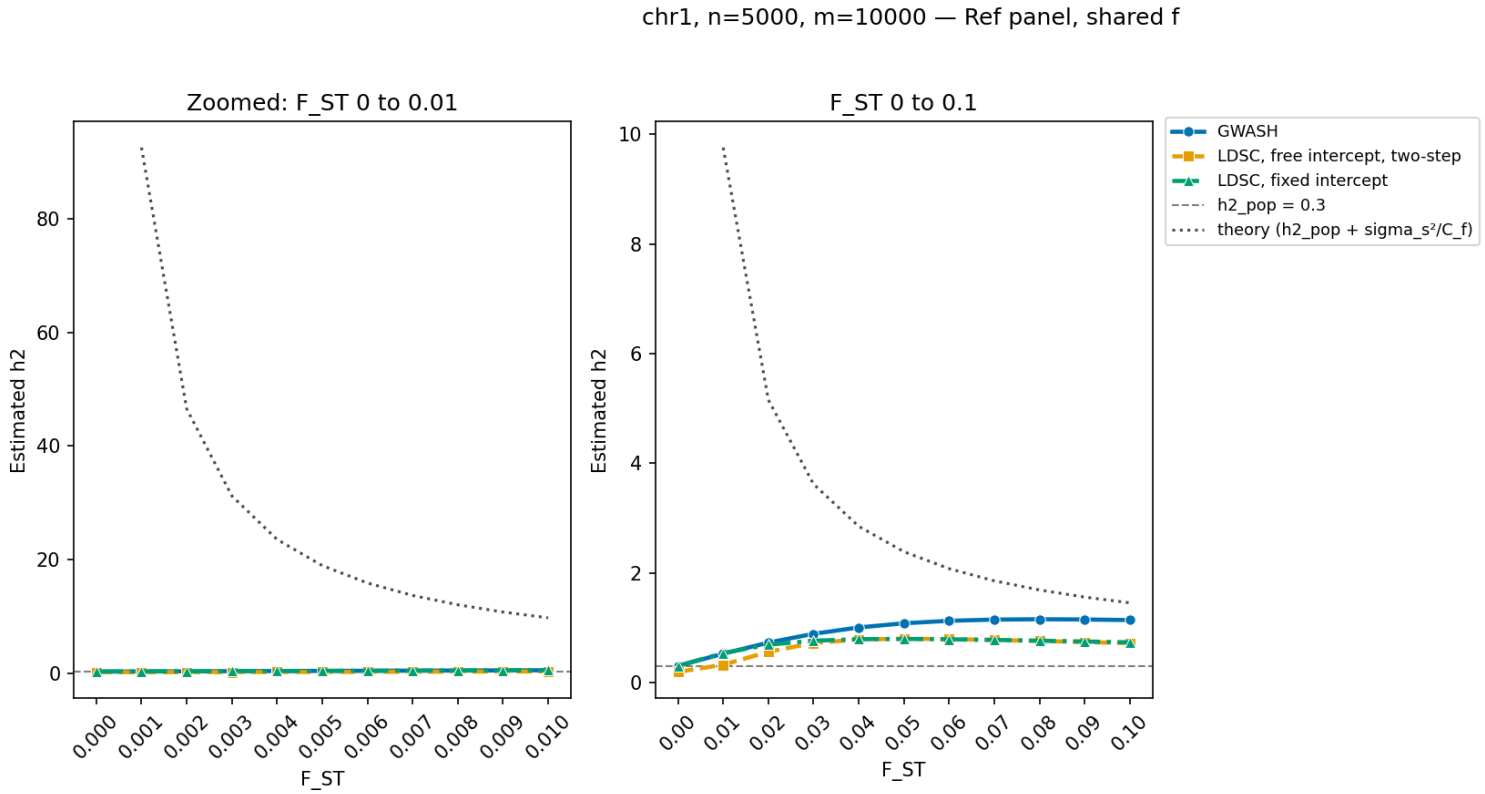
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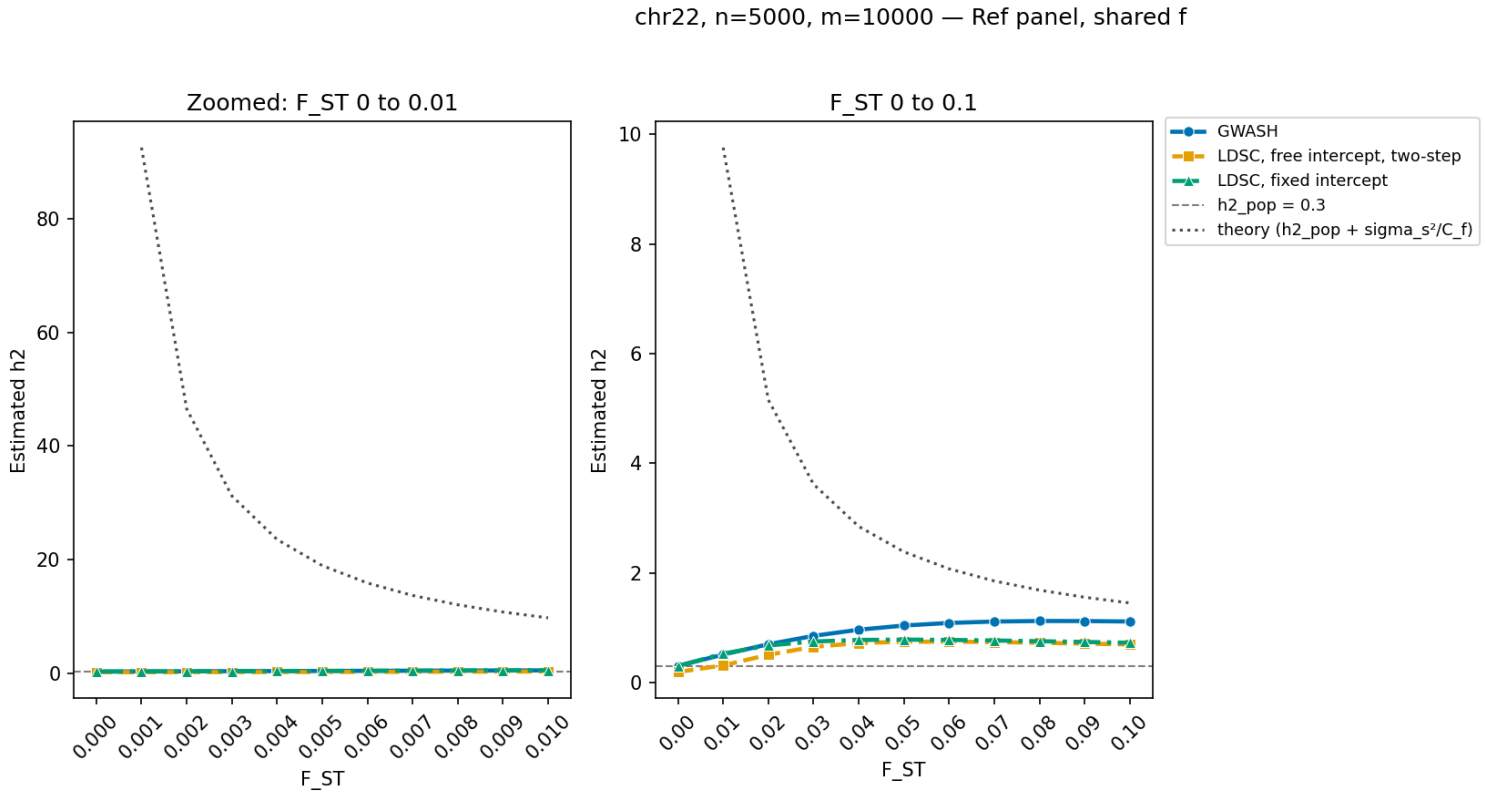
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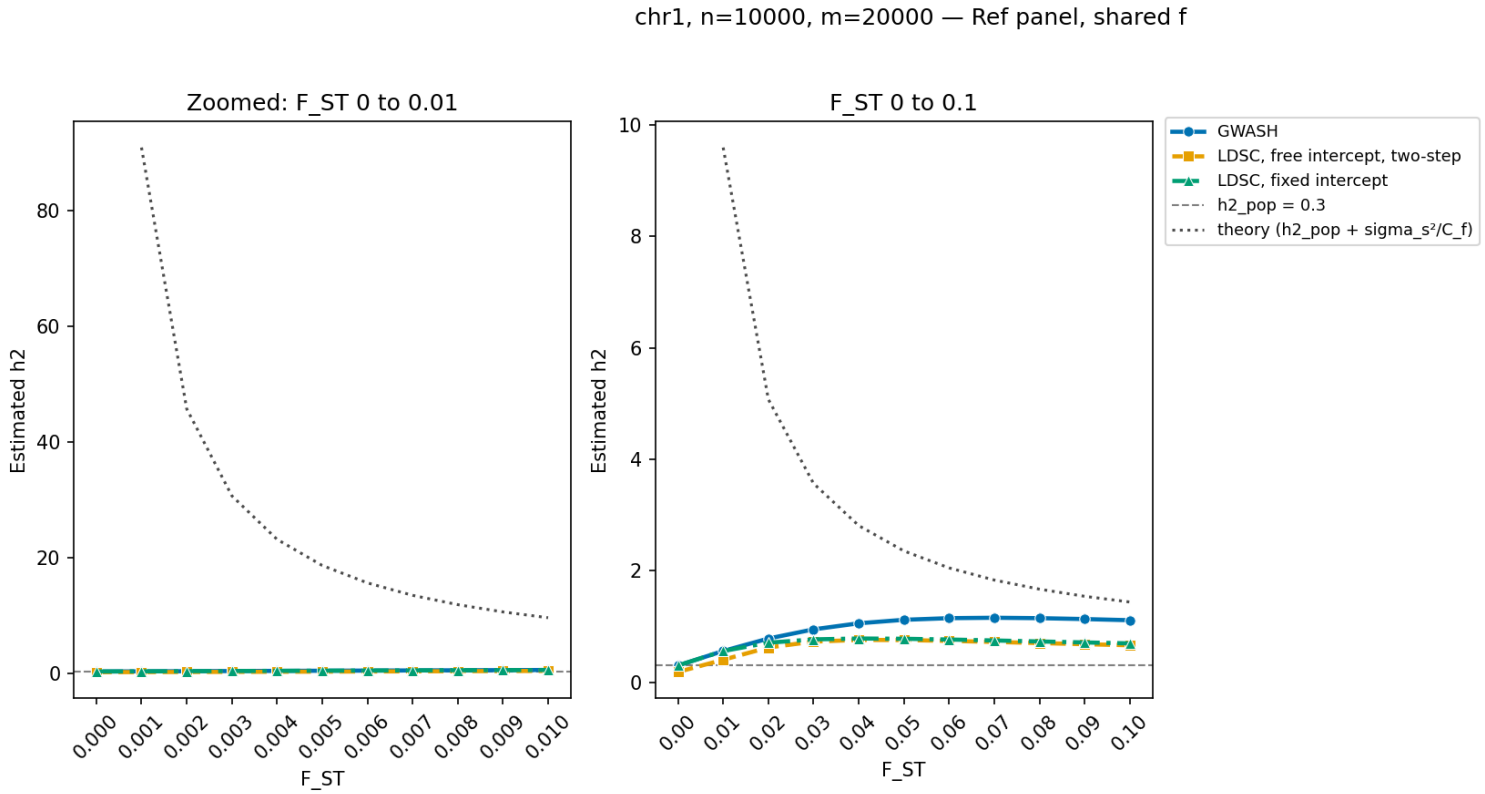
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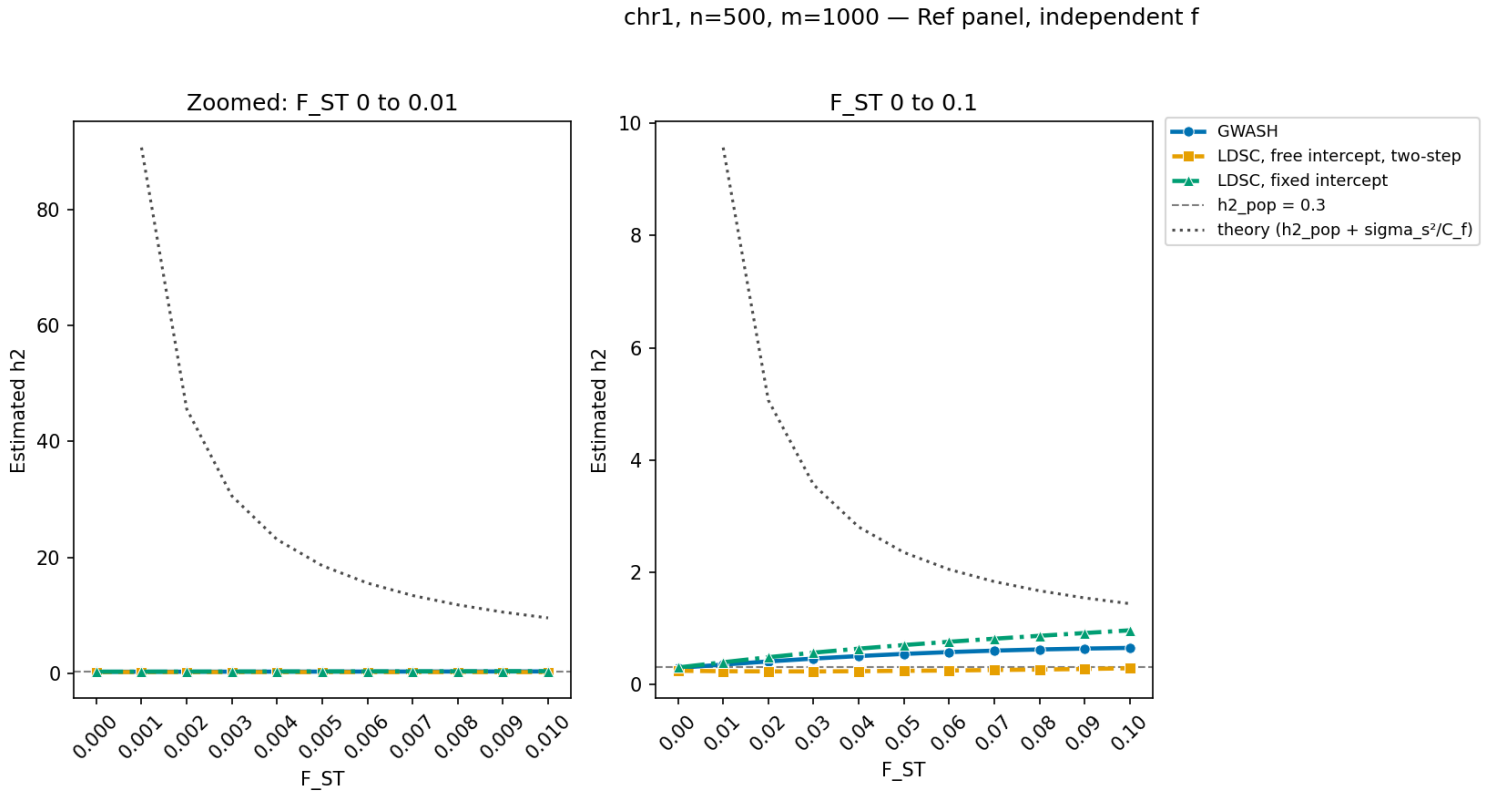
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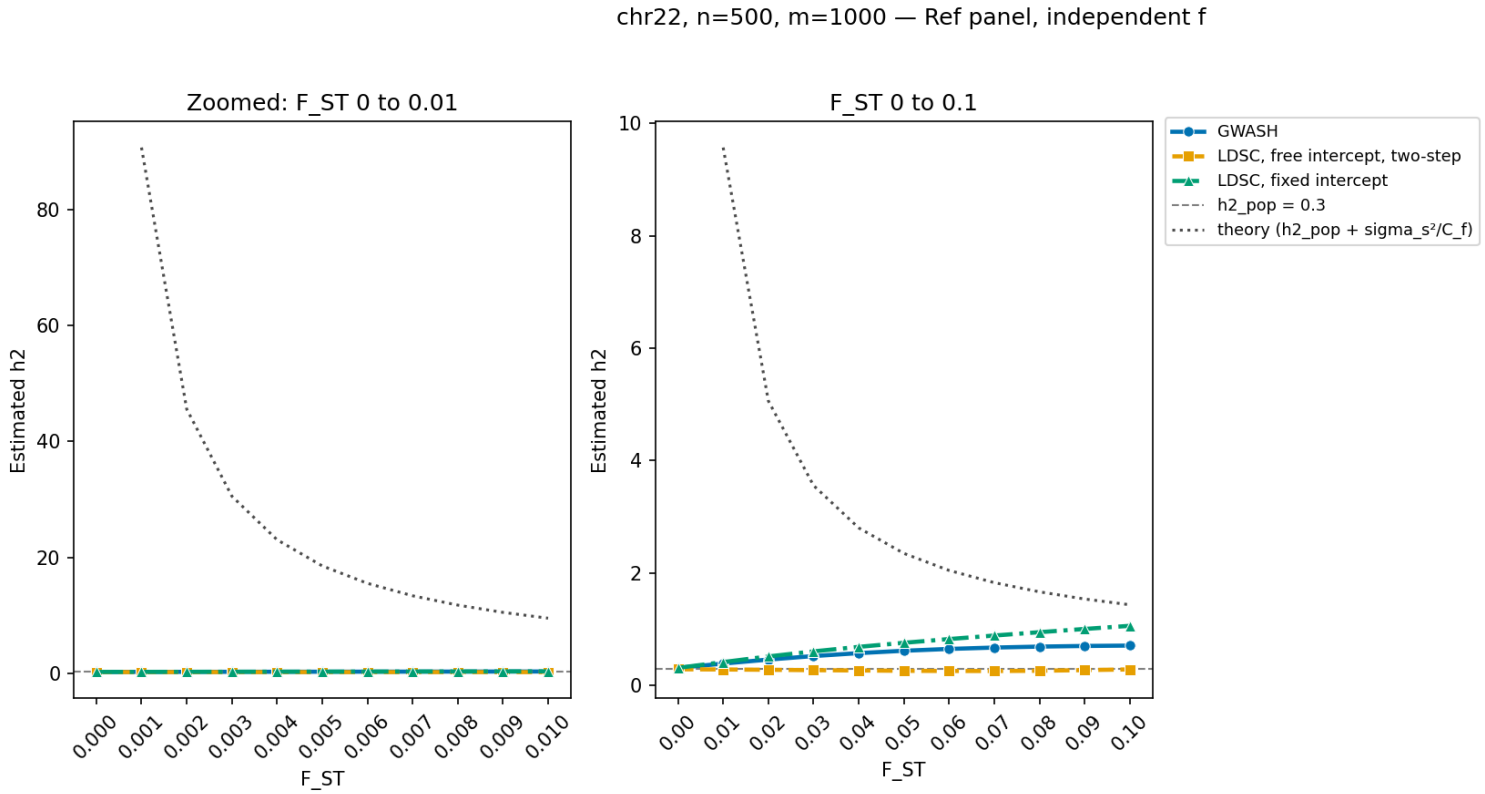
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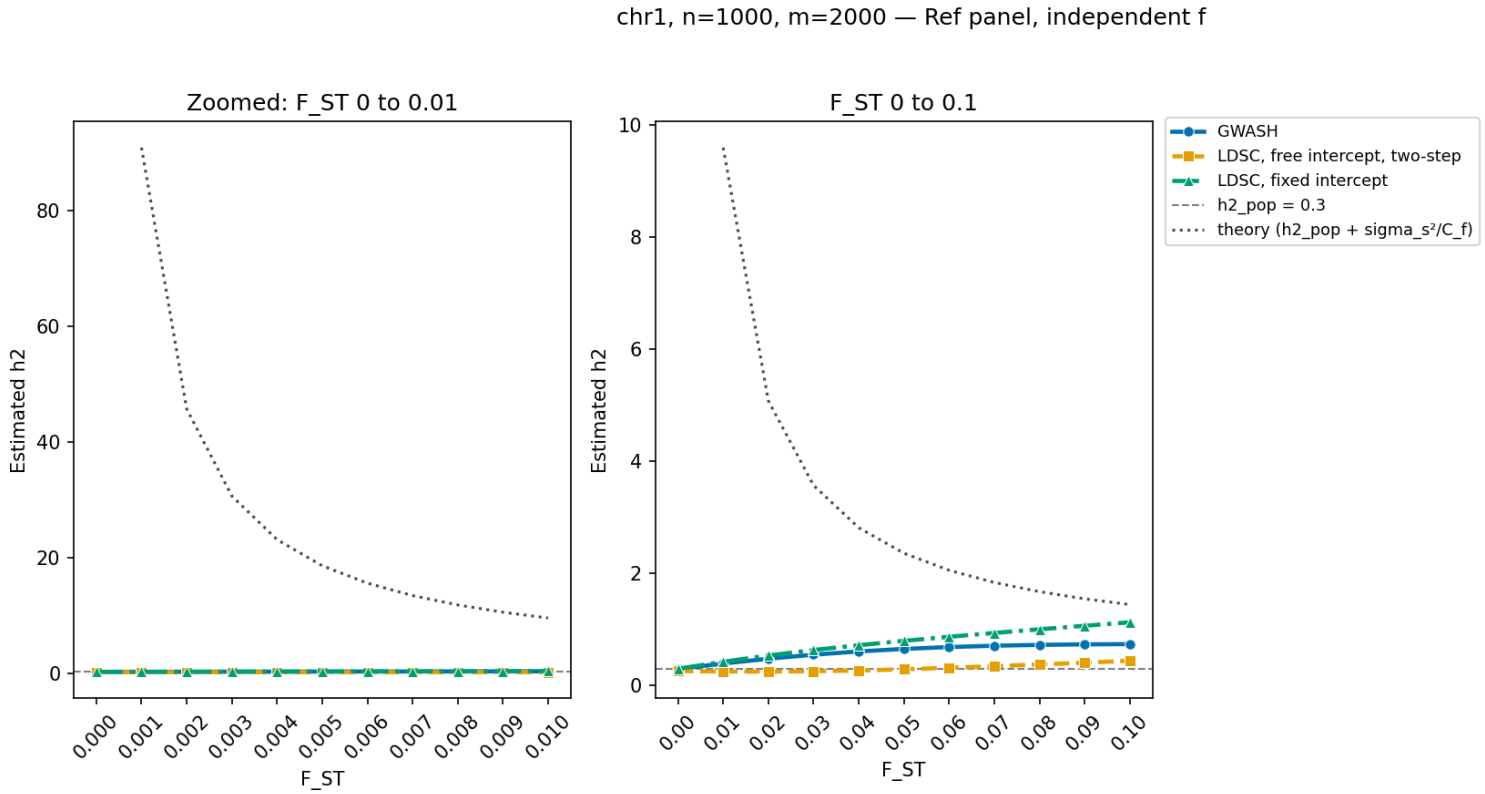
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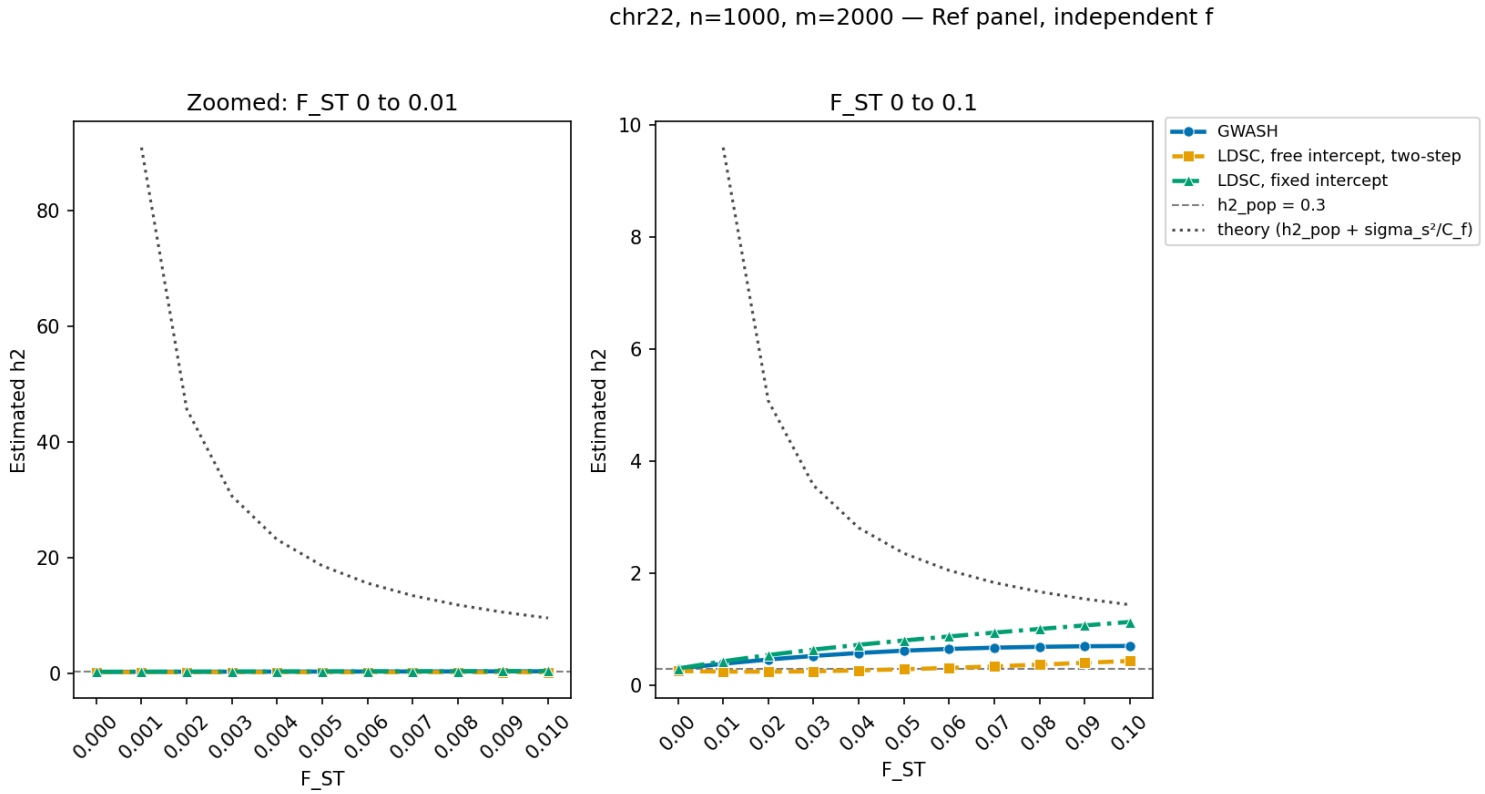
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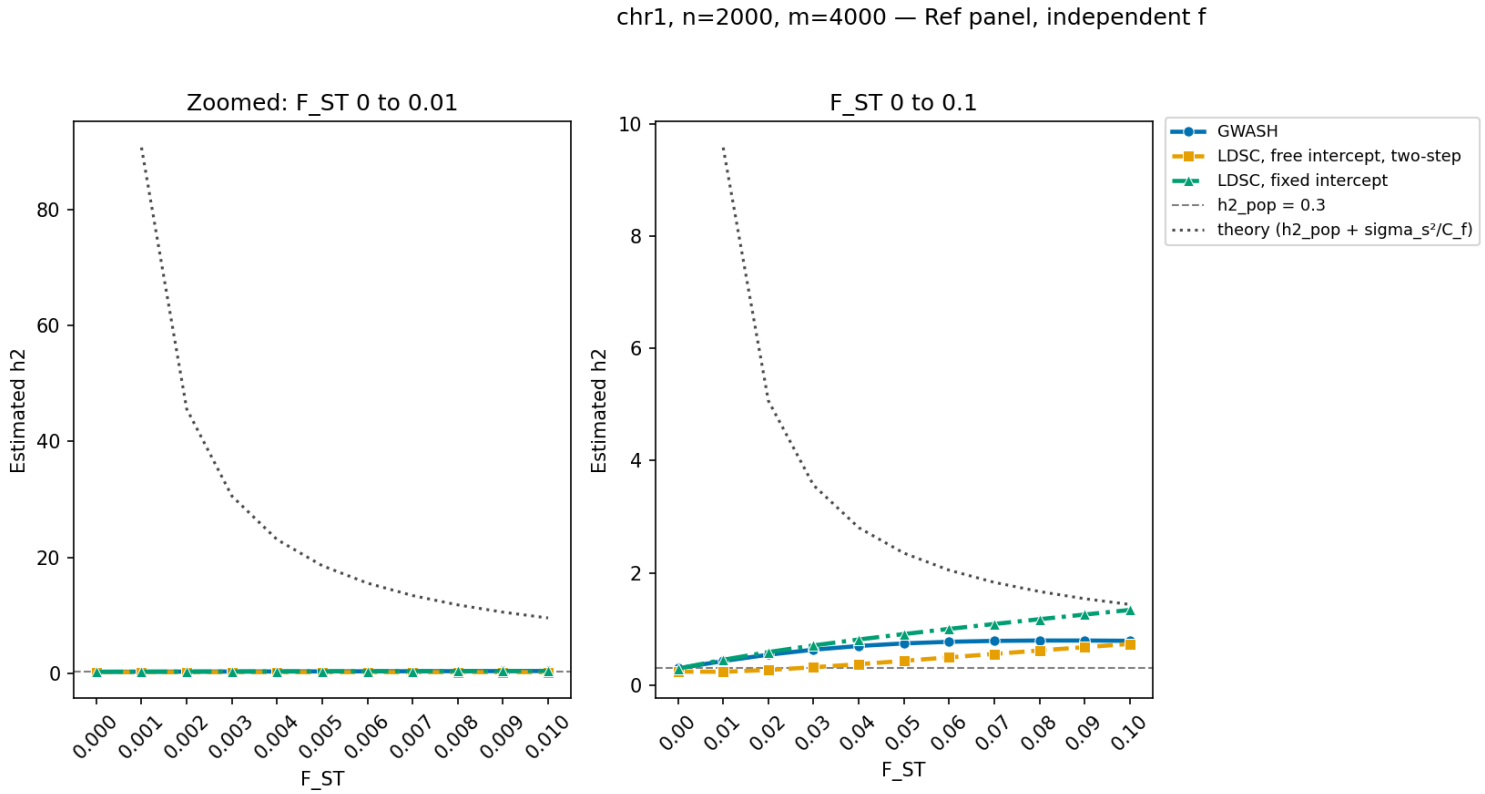
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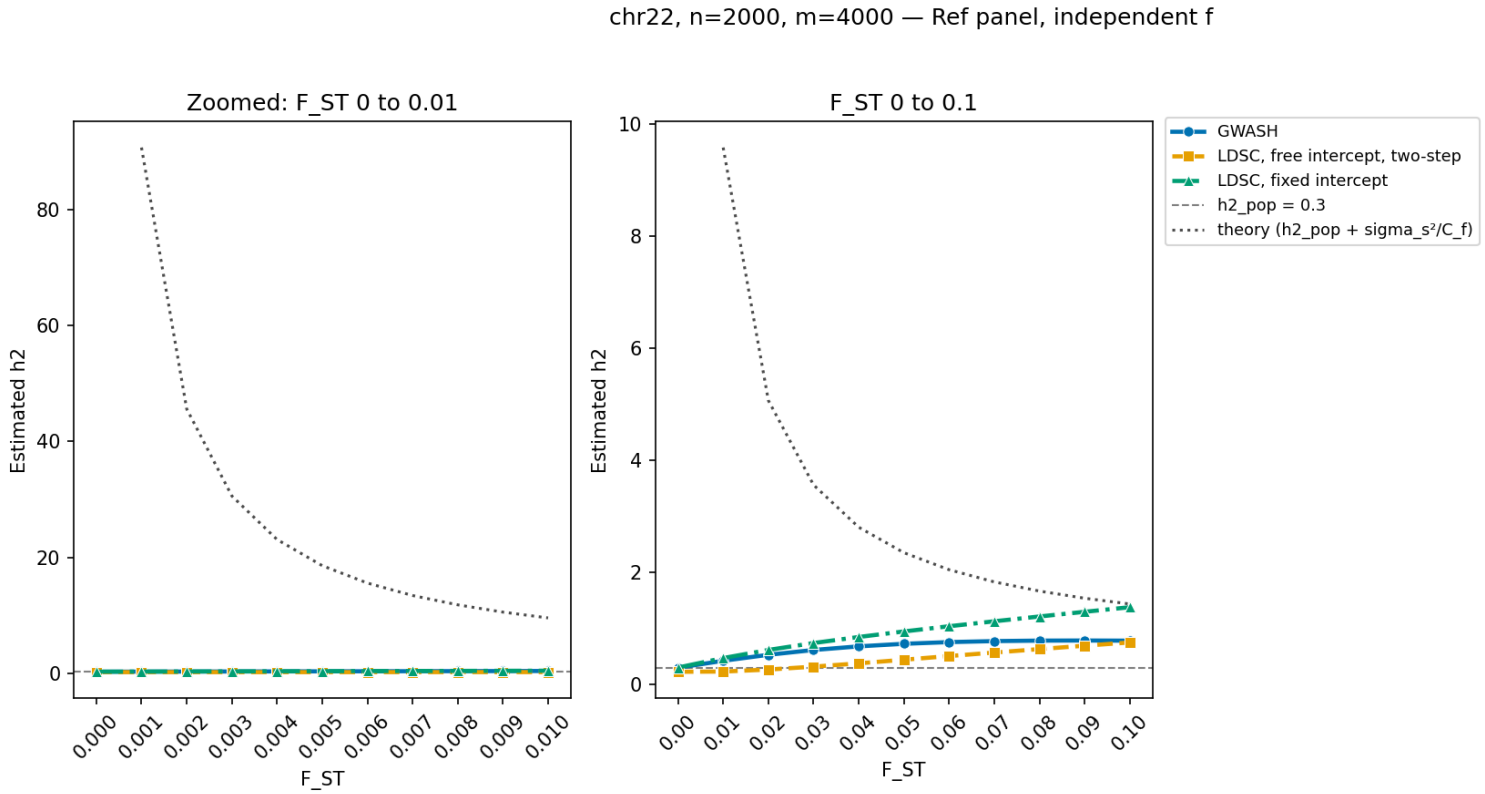
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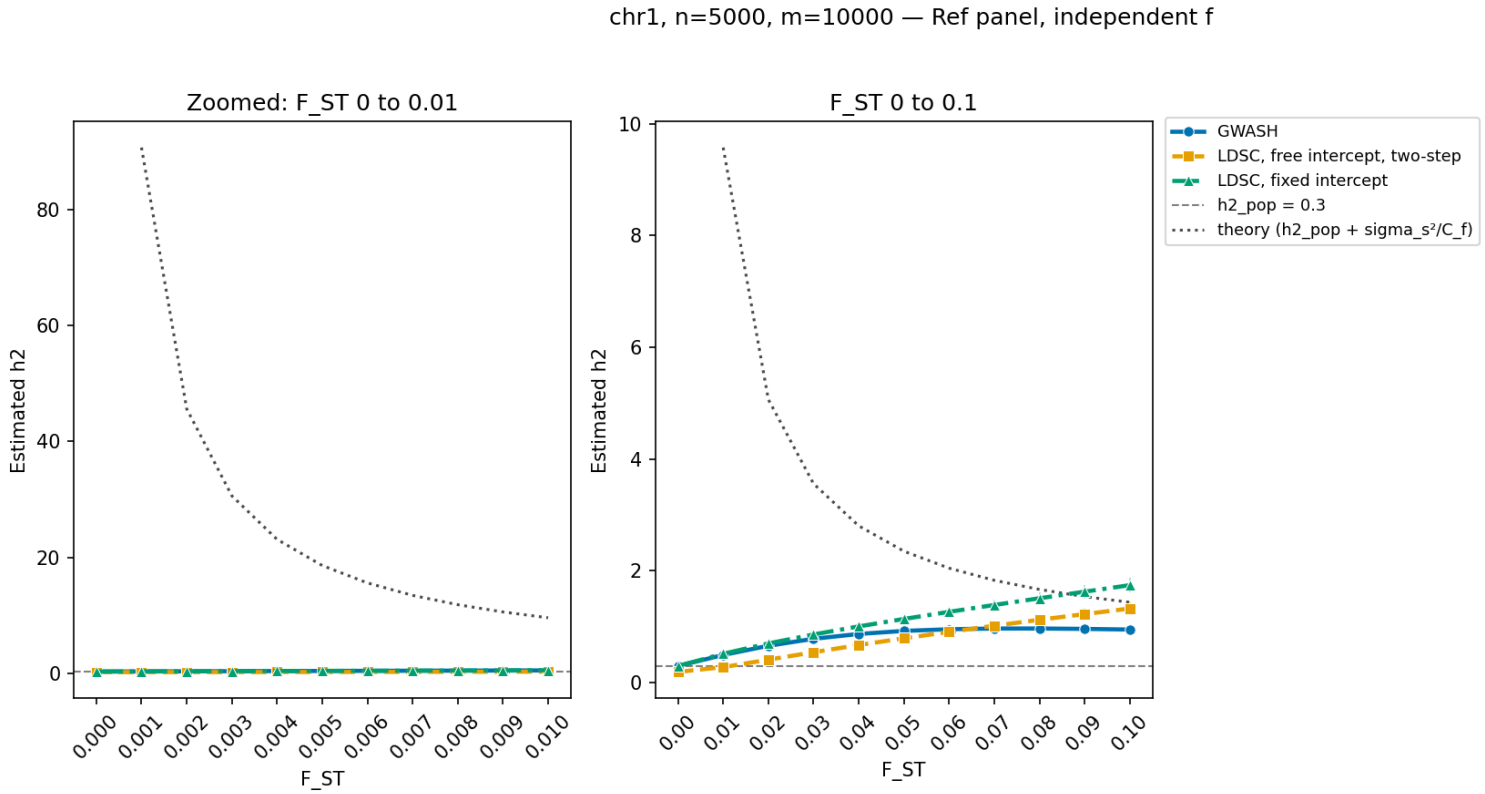
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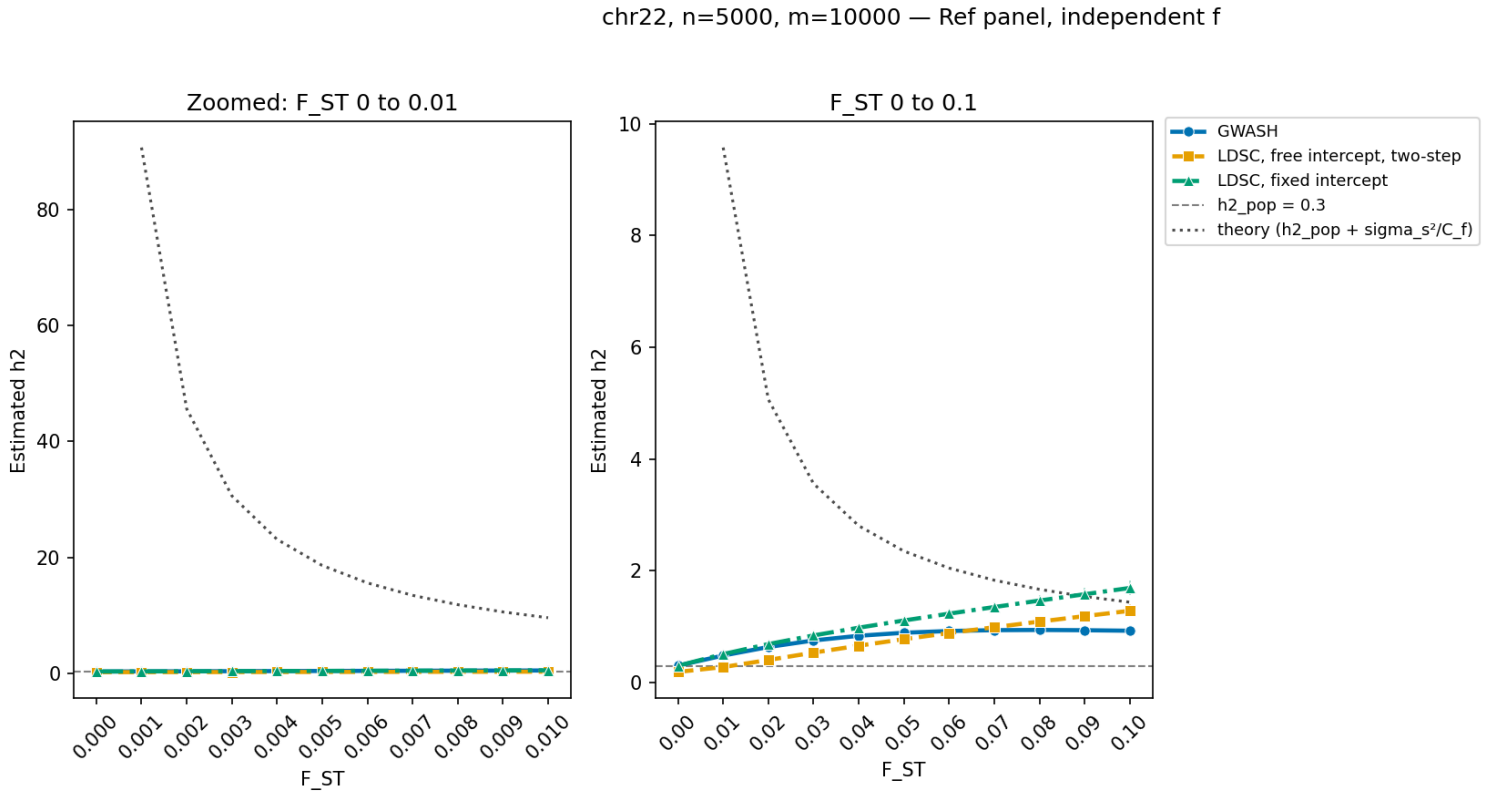
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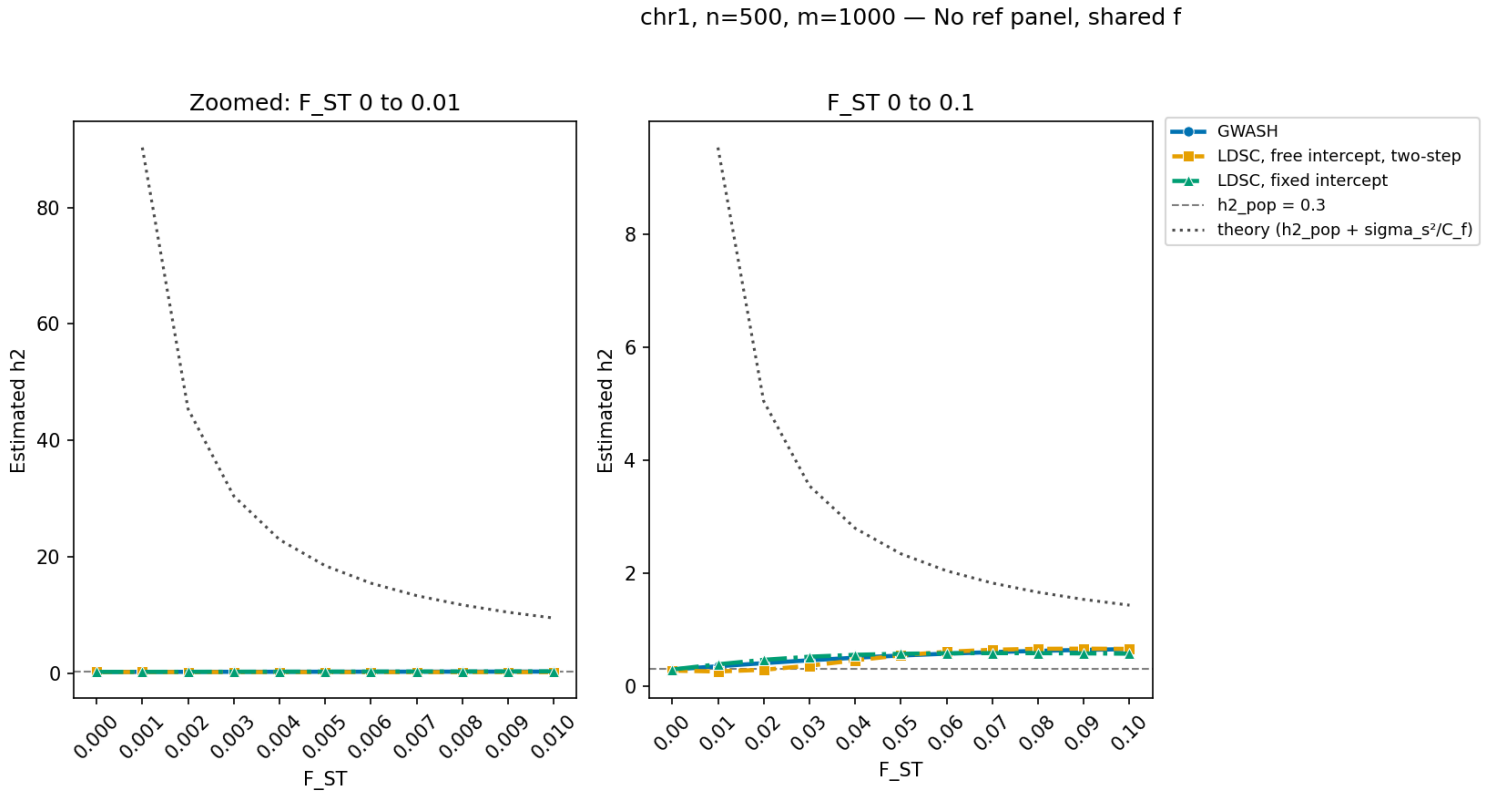
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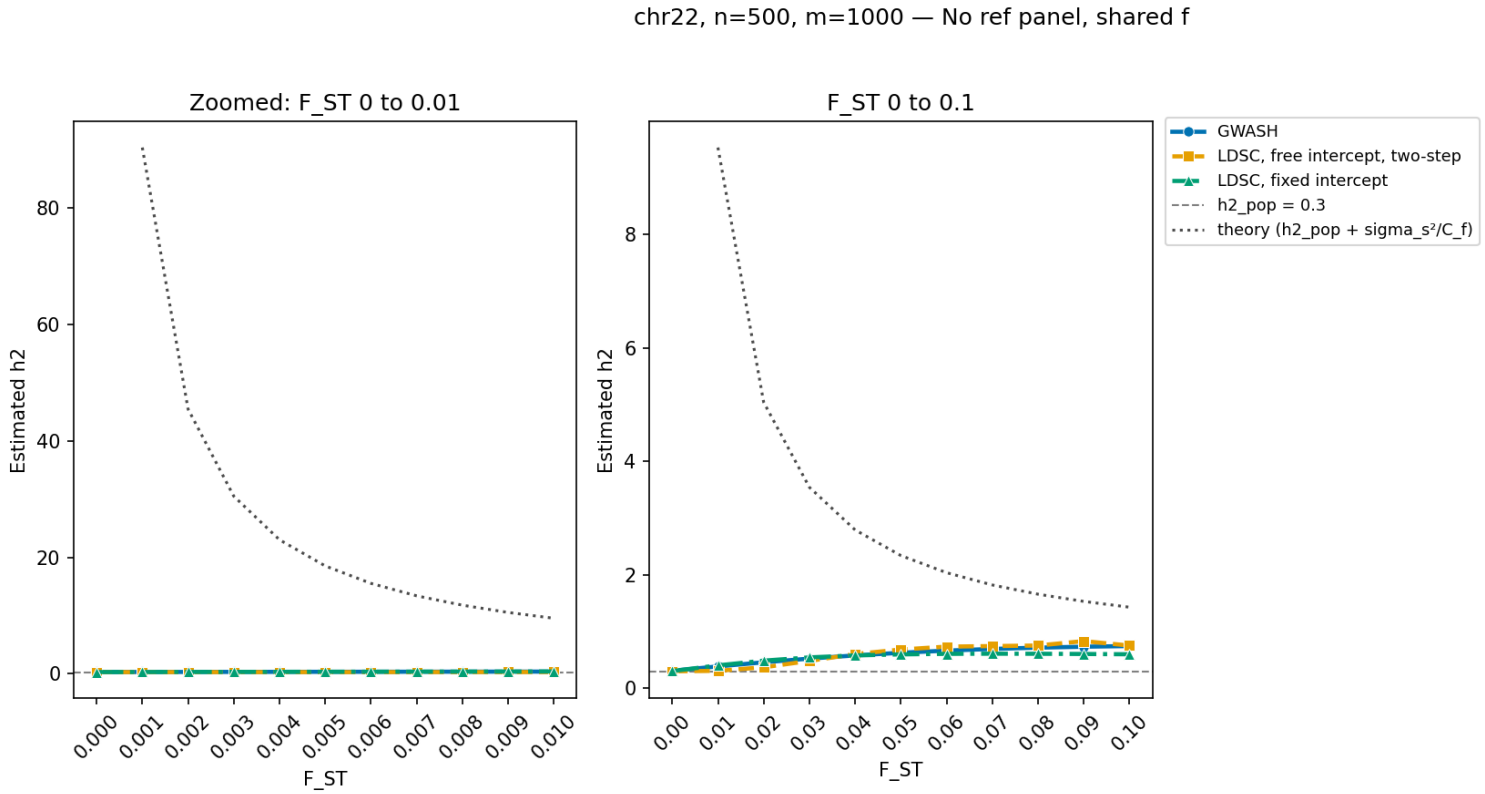
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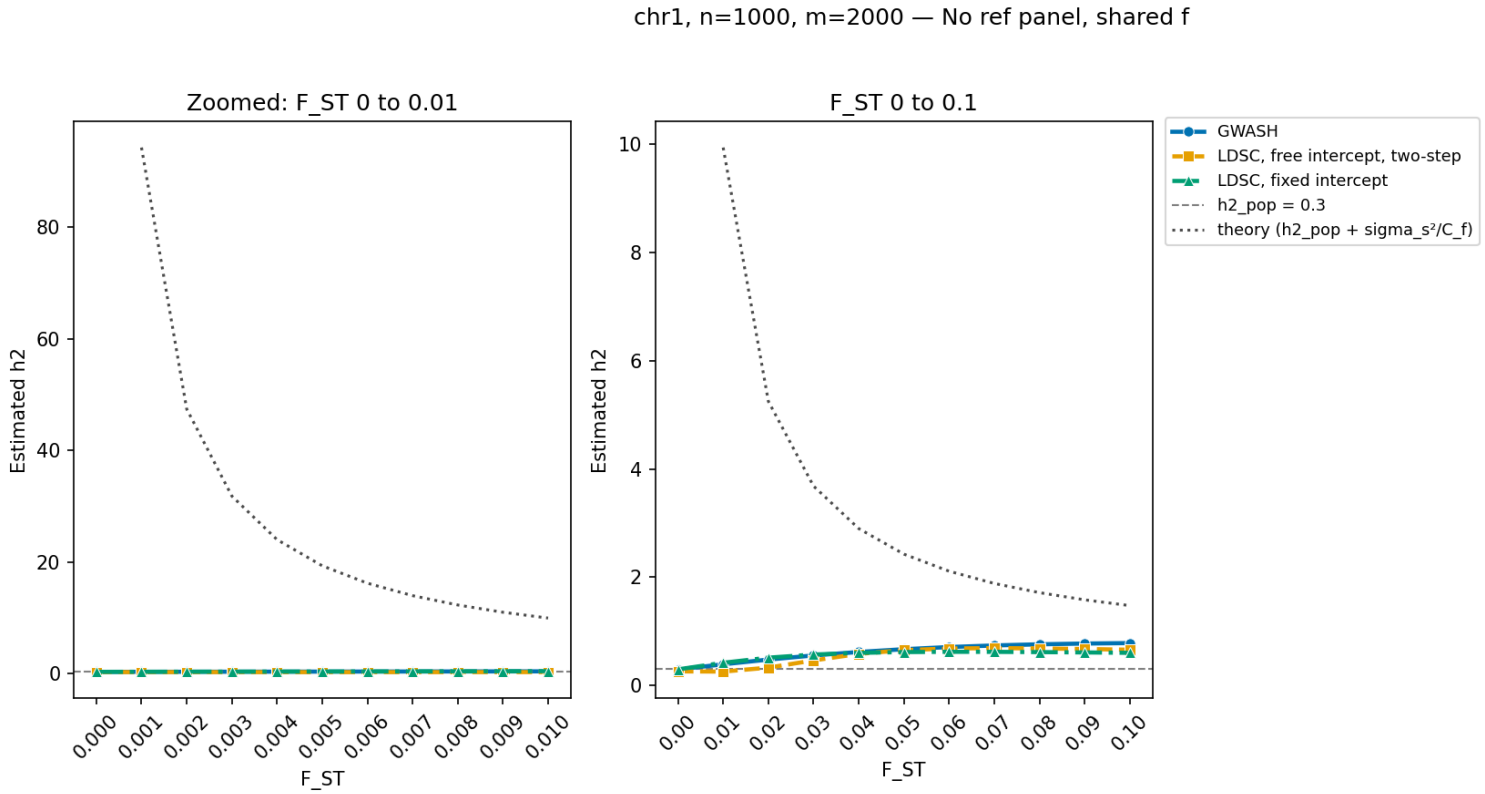
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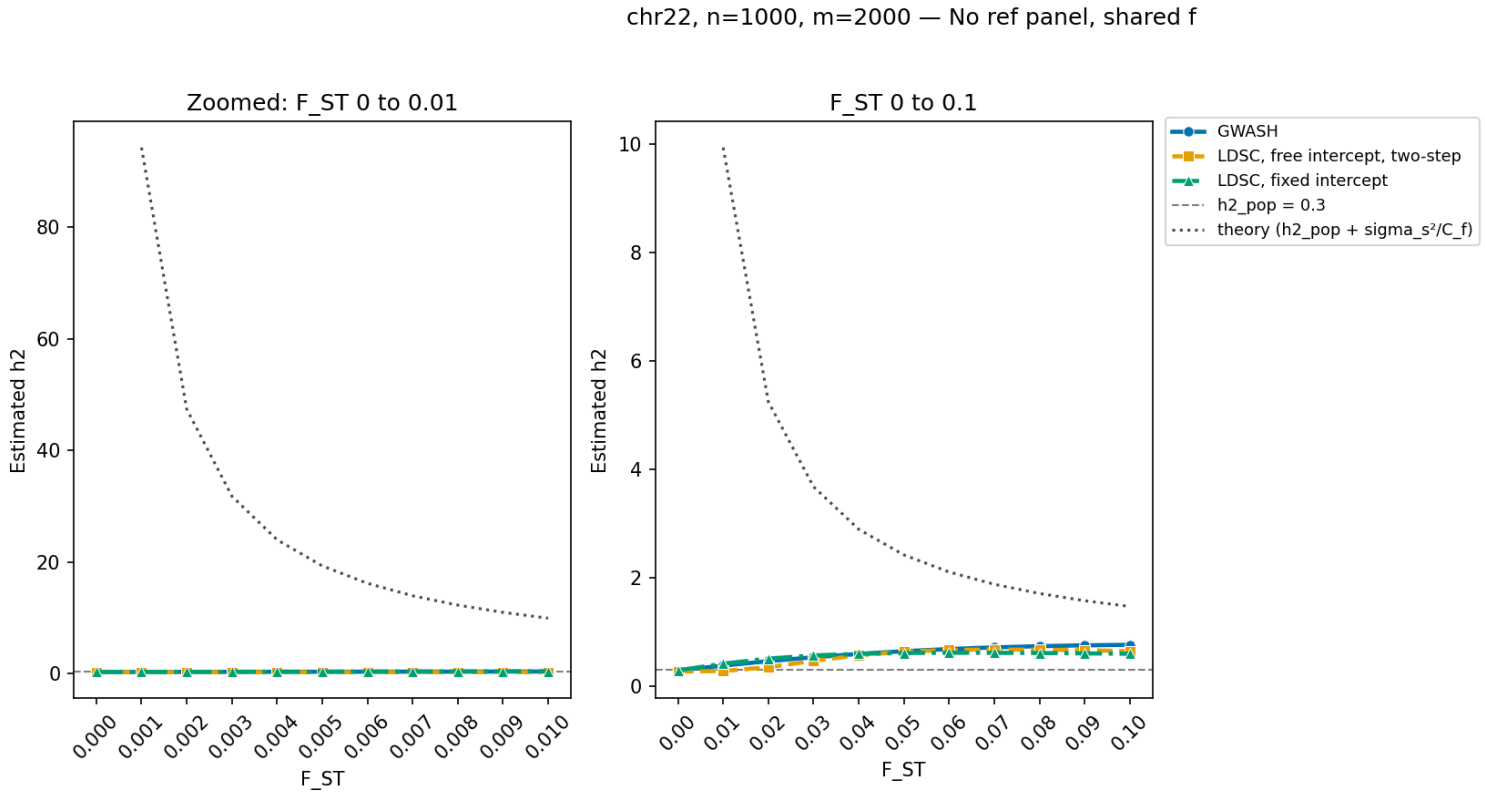
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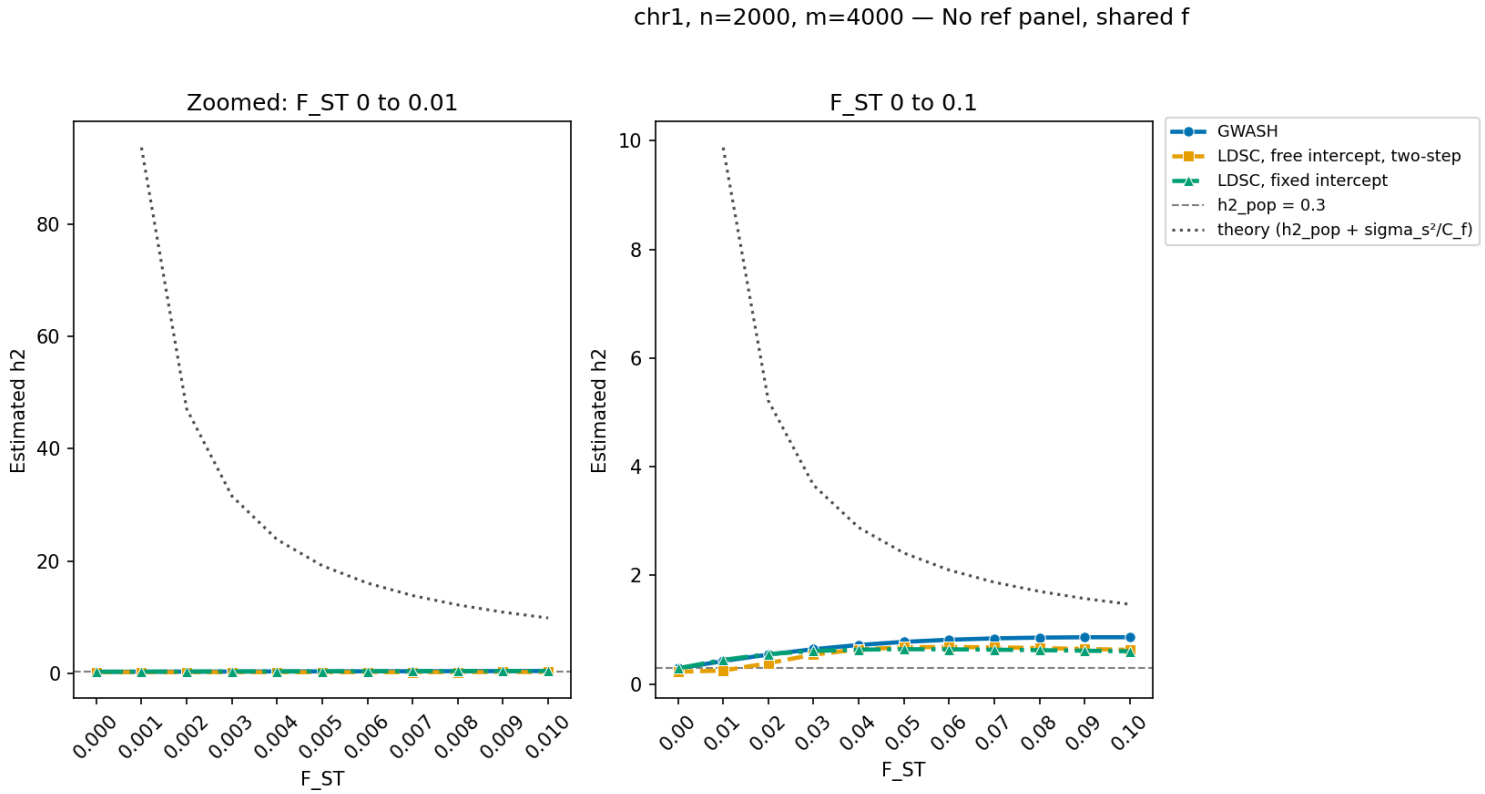
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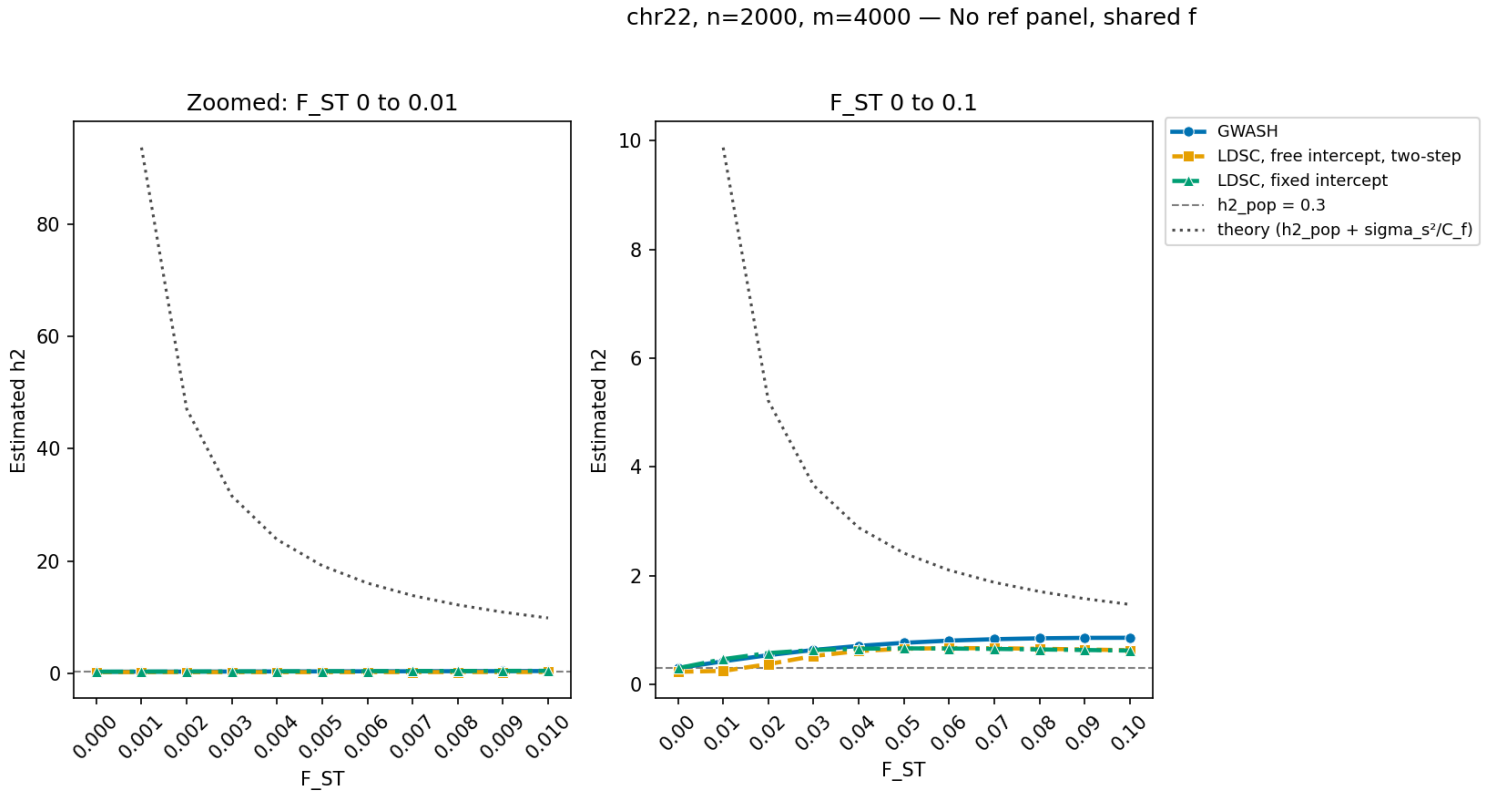
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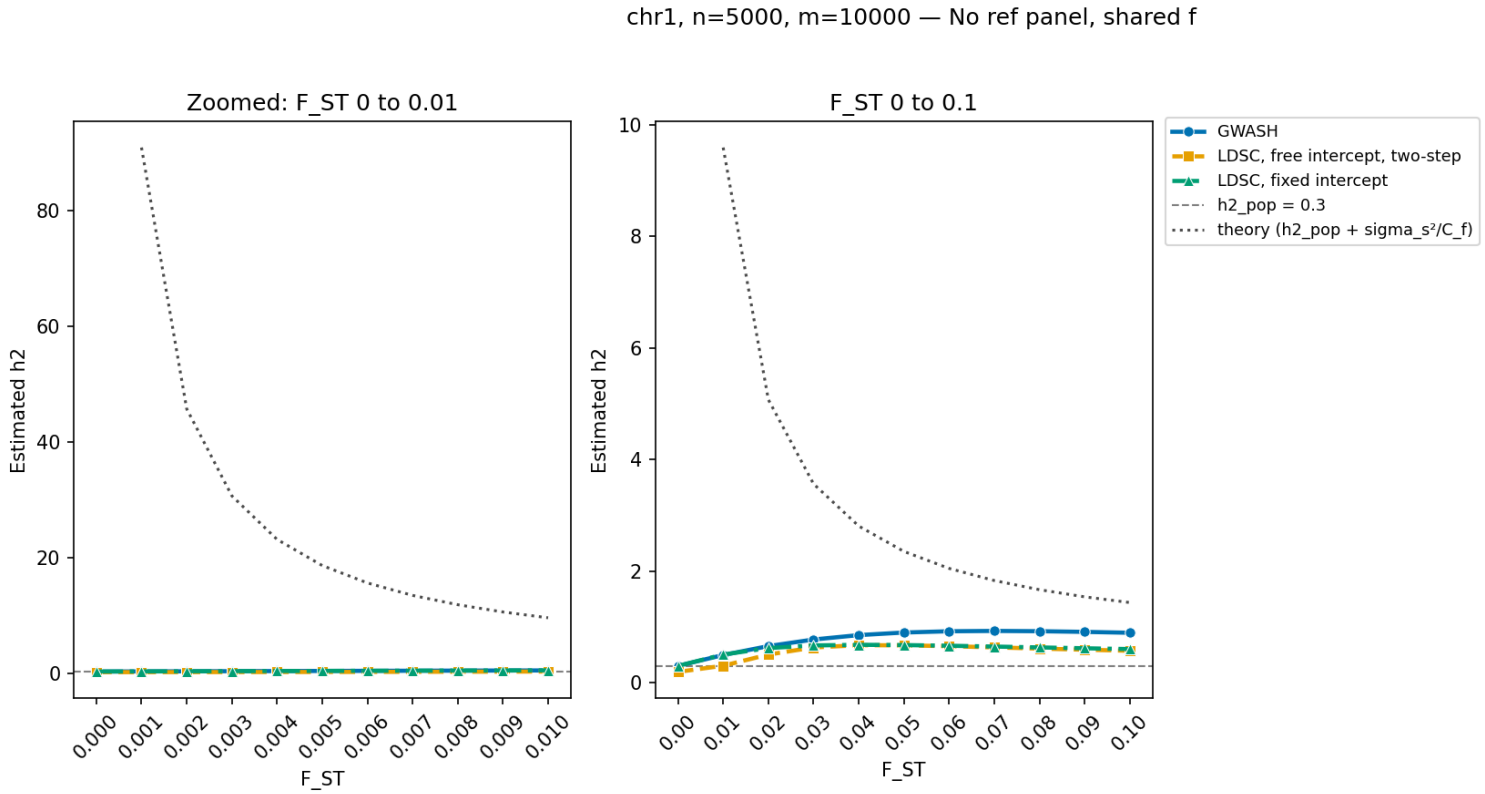
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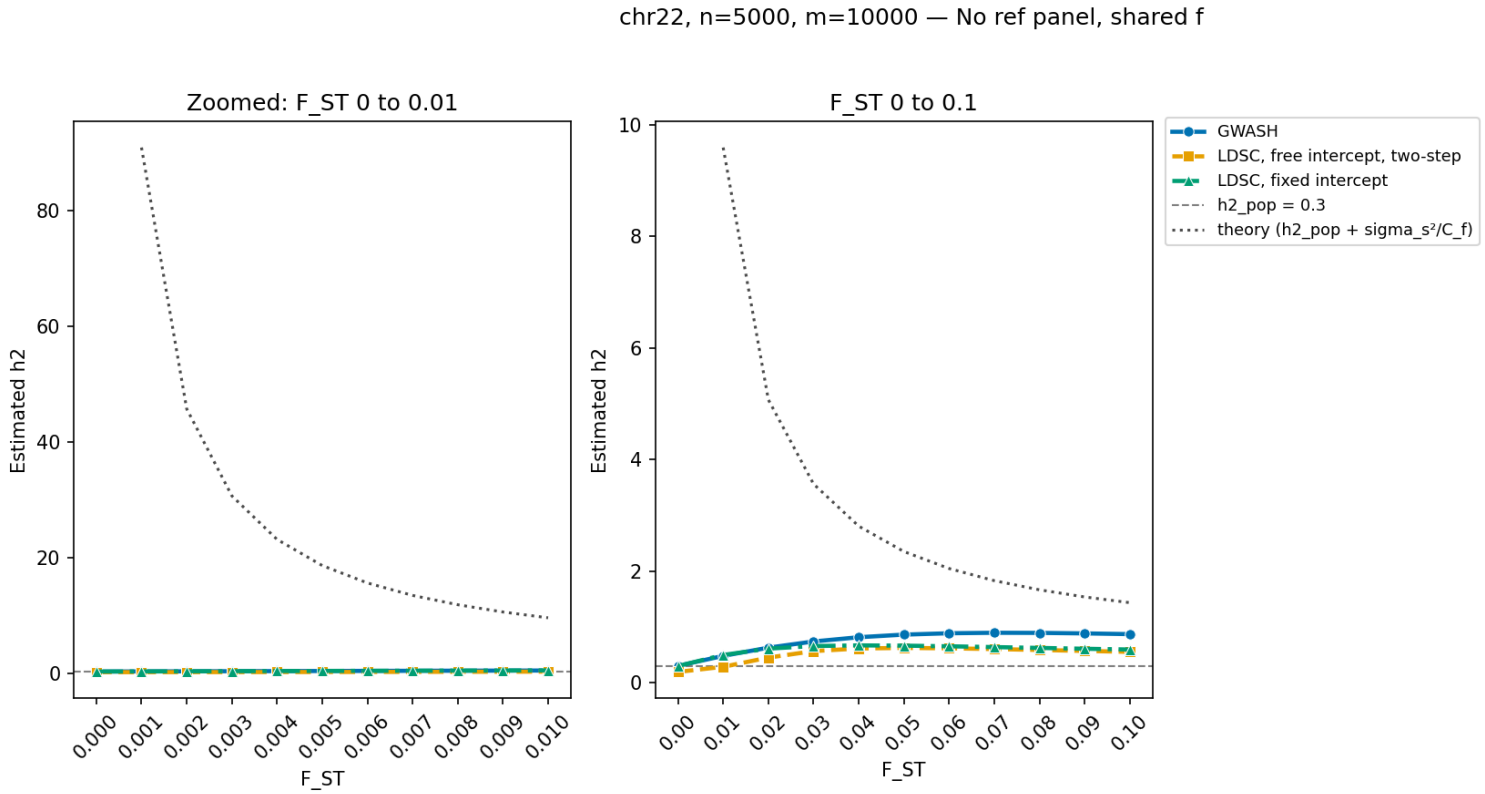
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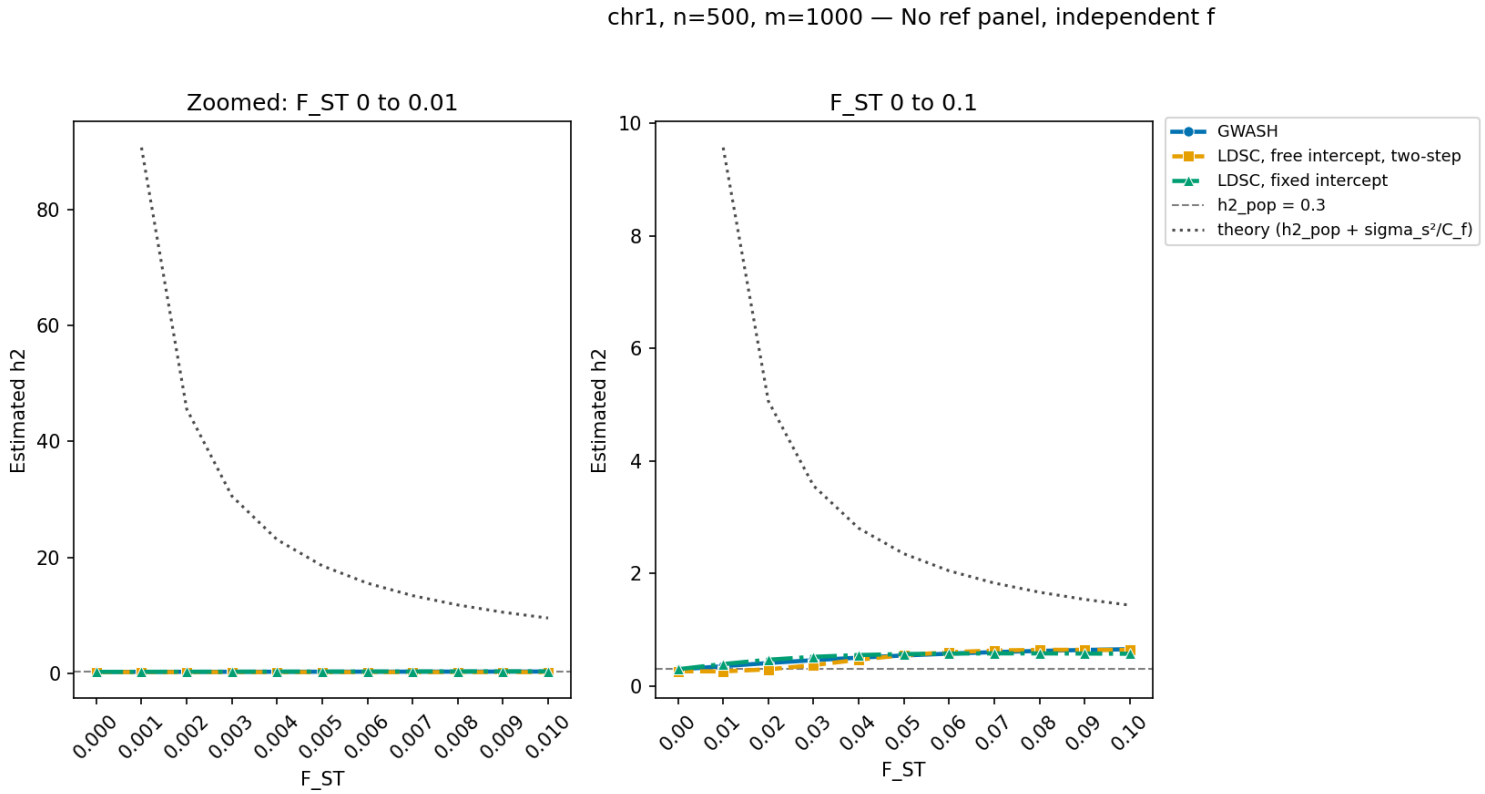
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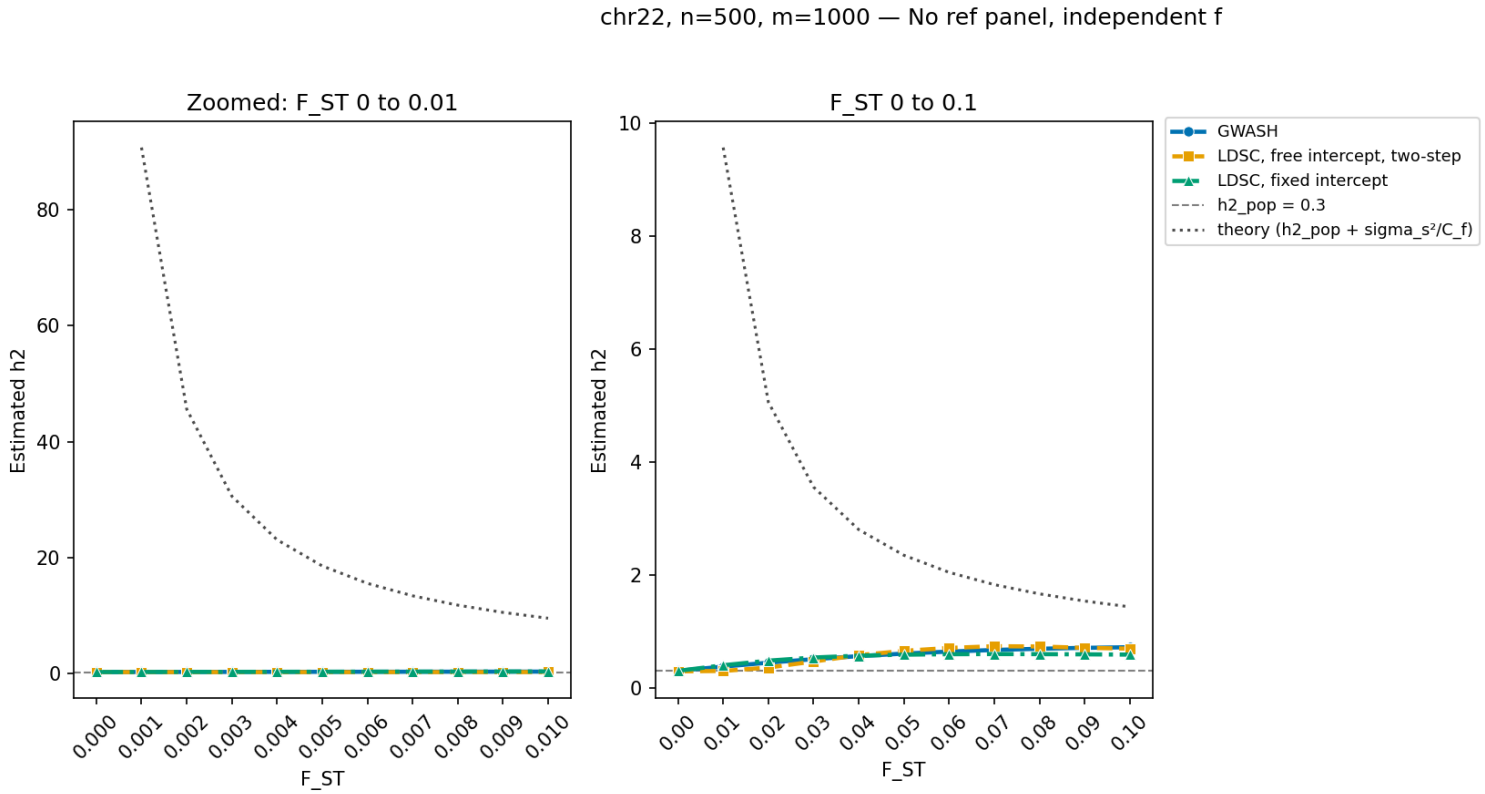
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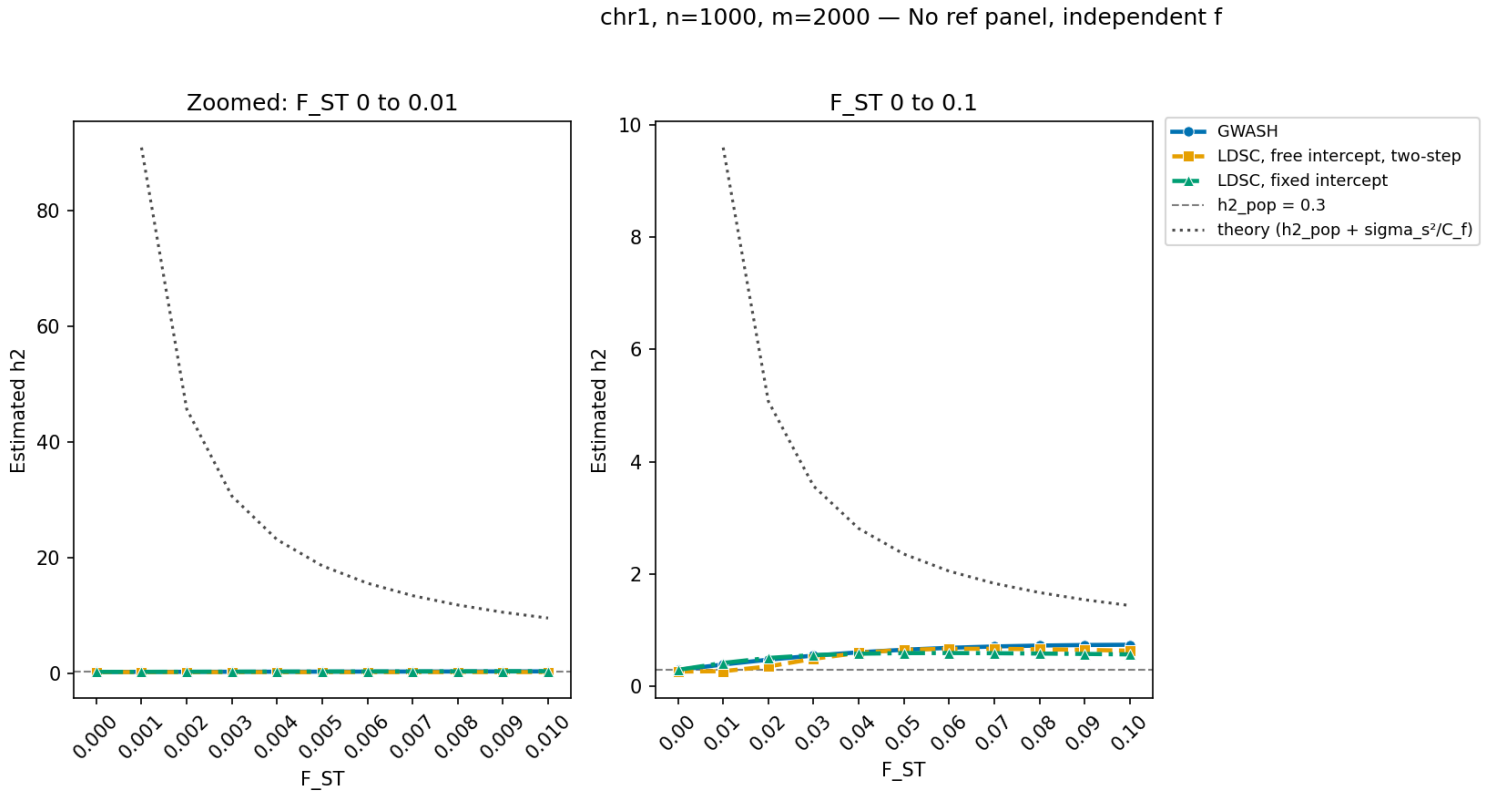
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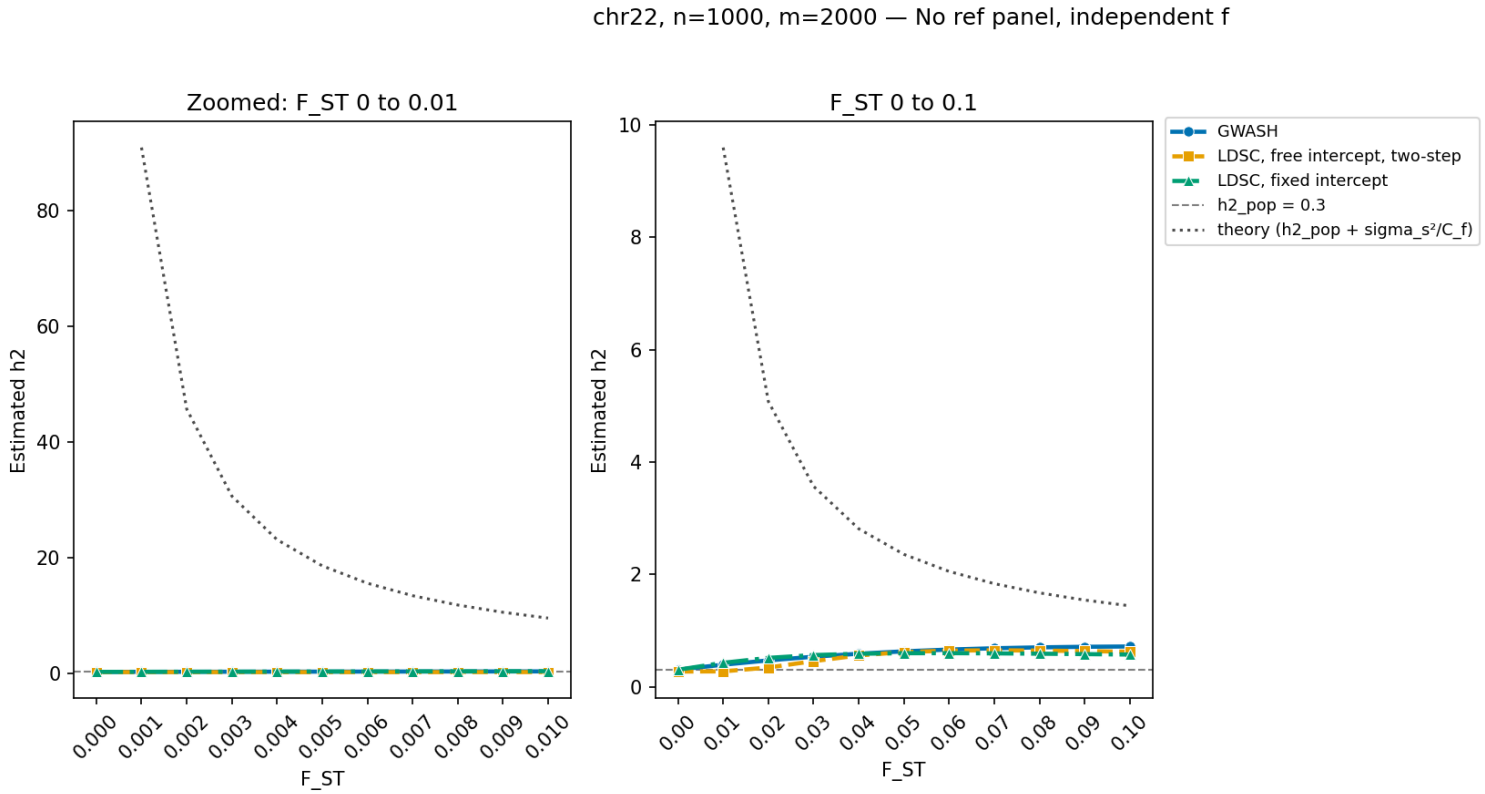
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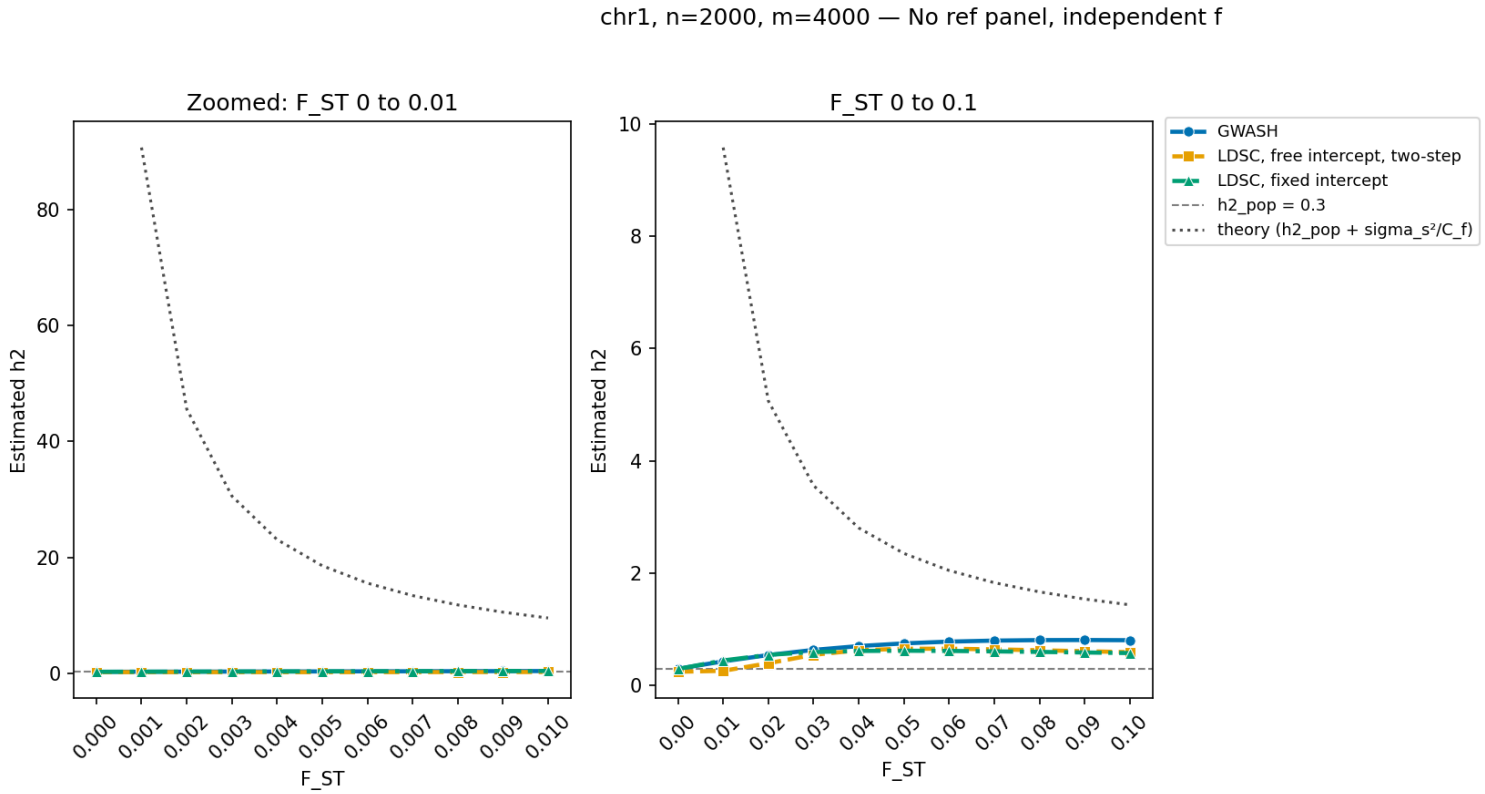
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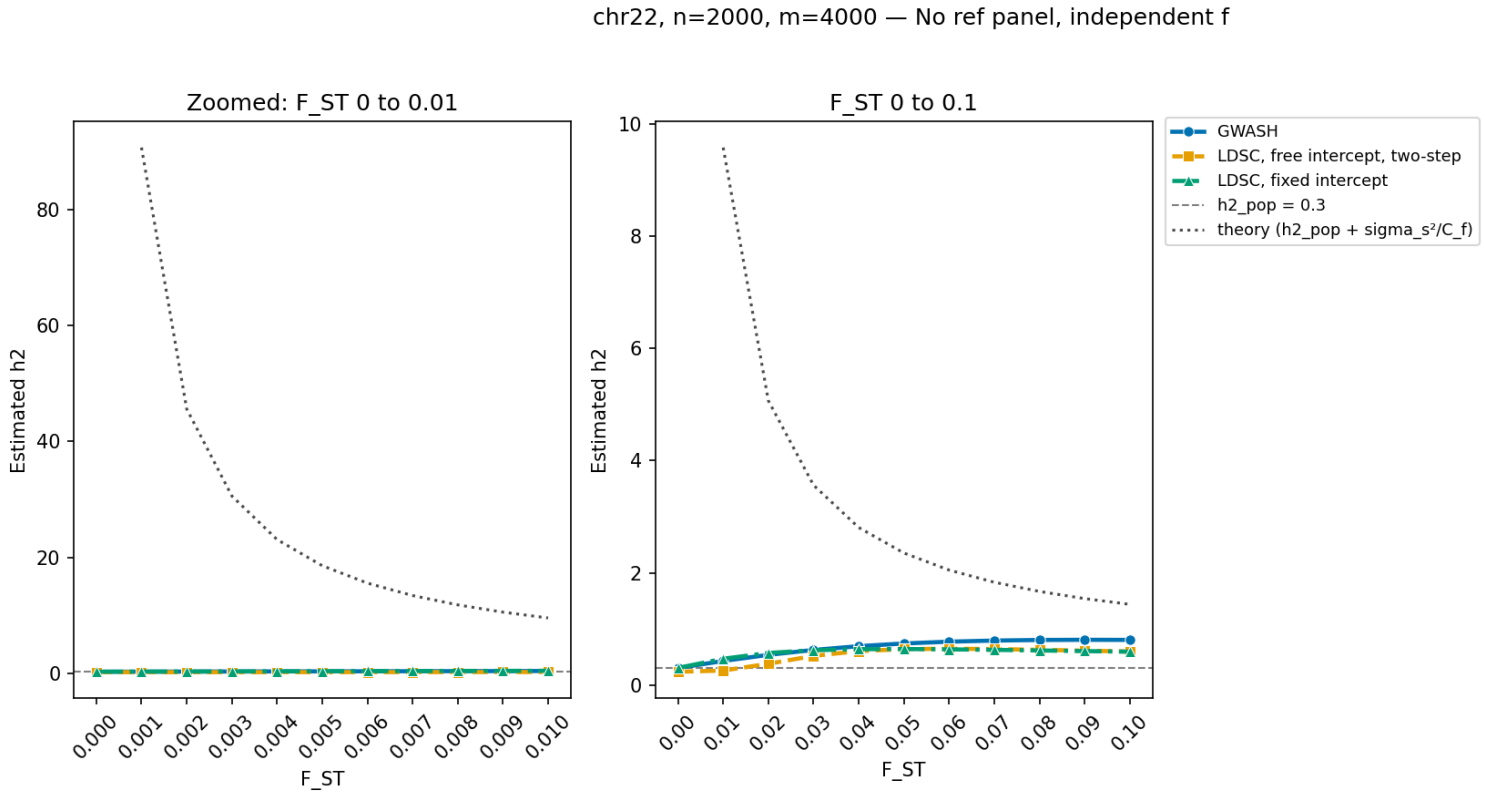
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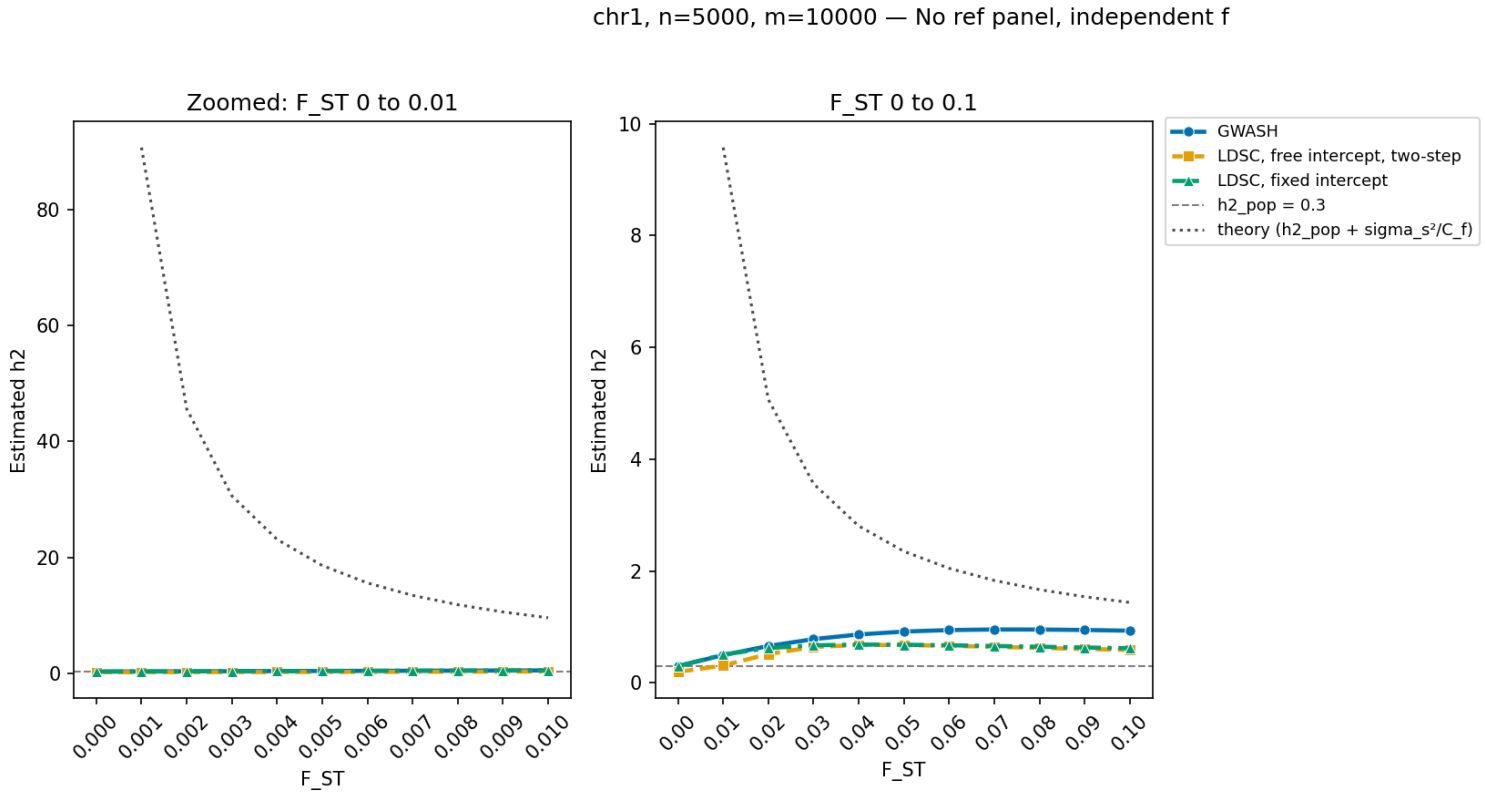
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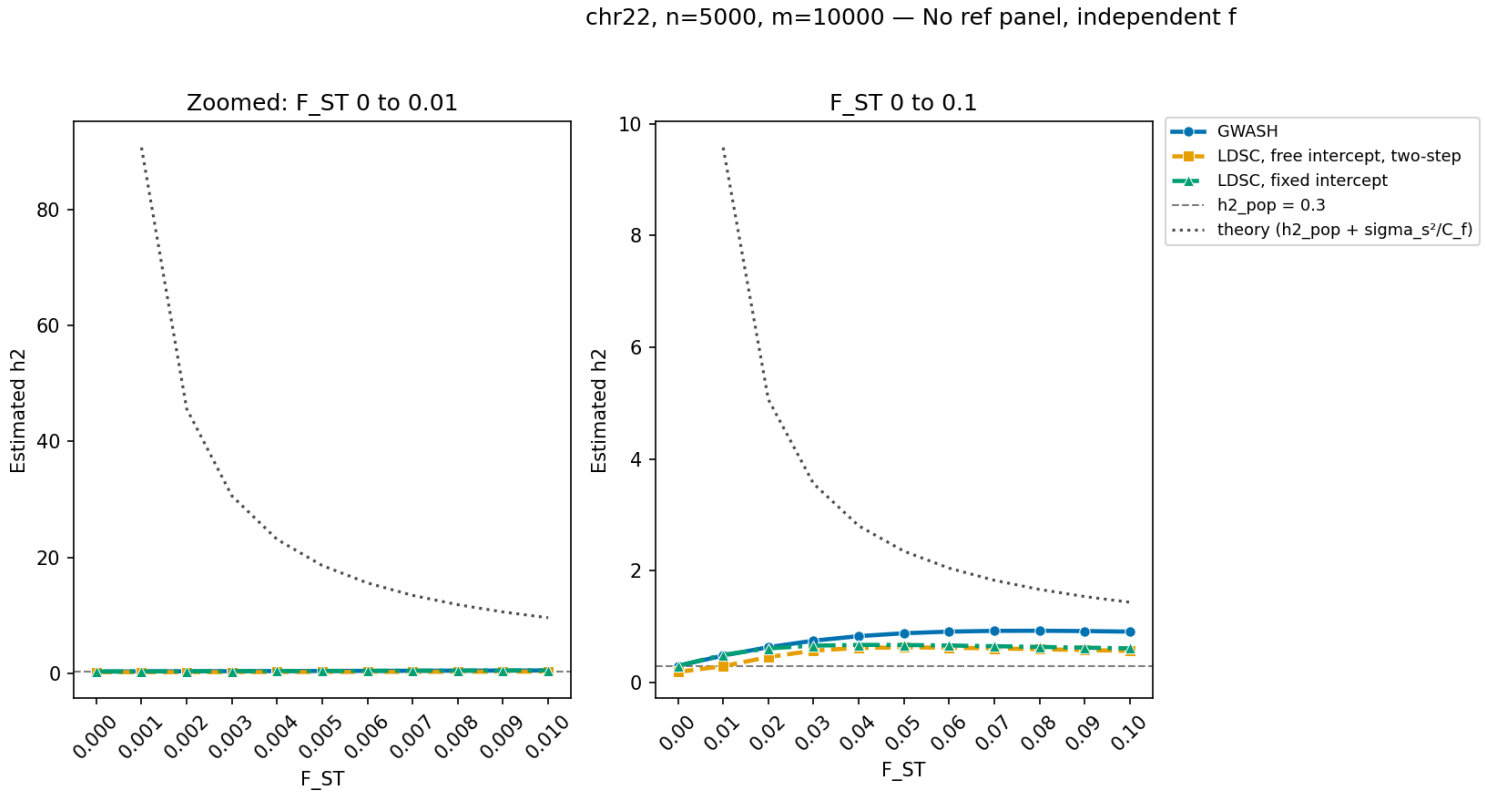
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# Discussion

**GWASH and LDSC agree when the reference panel is well-specified:** with no reference panel at all, or with a reference panel whose f matches the main sample, GWASH and both LDSC intercept choices move together with $F_{ST}$ and track the theory line - see the 'No ref panel' and 'Ref panel, shared f' figures below.

**They diverge once the reference panel is mismatched, but all three still grow with n:** in the 'Ref panel, independent f' scenario, LDSC's free-intercept estimate grows fastest with sample size (4.5-4.7× larger at n=5000 than at n=500, at $F_{ST}$=0.1), LDSC's fixed-intercept estimate grows more moderately (1.6-1.8×), and GWASH grows the least of the three (1.3-1.5×) over that same range of scales.

**Fixed vs. free intercept matters less than the reference-panel mismatch itself:** LDSC's fixed-intercept estimate is consistently larger than its free-intercept estimate in the mismatched scenario at small n (0.96 vs 0.28 at n=500), but the free-intercept estimate catches up and nearly matches it by n=5000 (1.33 vs 1.74) - both grow with scale, just starting from different magnitudes and at different rates.

**Caveat:** GWASH's own bias in the mismatched scenario (roughly 2-3× h2_pop within this $F_{ST}$ range, not 1×) means it is not immune to the f-mismatch - it is simply less sensitive to it than LDSC's free-intercept estimate, and its growth with sample size is slower than either LDSC intercept choice. This report does not investigate why GWASH's bias grows more slowly with n.